Science Robotics
AAAS

Supplementary Materials for

Swarm of micro flying robots in the wild

Xin Zhou *et al.*

Corresponding author: Fei Gao, fgaoaa@zju.edu.cn; Chao Xu, cxu@zju.edu.cn

Sci. Robot. **7**, eabm5954 (2022)
DOI: 10.1126/scirobotics.abm5954

The PDF file includes:

Sections S1 to S10
Figs. S1 to S17
Tables S1 to S13

Other Supplementary Material for this manuscript includes the following:

Software

Science Robotics
AAAS

补充材料

野外微型飞行机器人群

周欣 等人

通讯作者：高飞，fgaoaa@zju.edu.cn；徐超，cxu@zju.edu.cn

科学机器人 **7**，eabm5954 (2022)
DOI: 10.1126/scirobotics.abm5954

PDF文件包括：

第S1节至S10节 图
S1至S17 表S1至S13

本手稿的其他补充材料包括以下内容

软件

Section S1. Extra Comparison of Reciprocal Collision Avoidance

Table S1. Extra performance comparison. Comparison is performed under a position swap configuration in an 8×8 m empty space containing eight agents with radius of 0.25 m. Maximum velocity and acceleration are set to 1.7 m/s and 6 m/s^2, respectively. **Min. Clear.** denotes the minimum distance from collision. **Smoothness** is computed from the time integral of squared jerk $\int ||j||_2^2 dt$. Units of time and distance are second and meter, respectively.

Online/ Offline	Method	Comp. Time	Traj. Time	Traj. Length	Min. Clear.	Safety	Smoothness
Offline	SCP, h_scp=0.3s	1.46	7.65	**<u>9.64</u>**	-0.37	No	7.01
	SCP, h_scp=0.17s	7.72	**<u>7.40</u>**	9.70	-0.26	No	17.5
	RBP,no batches	**0.320**	15.4	11.3	0.01	Yes	0.608
	RBP,batch_size=4	0.413	15.4	11.5	0.03	Yes	**<u>0.604</u>**
Online	Mader	0.0239	**7.50**	12.2	0.19	Yes	125.1
	EGO, horizon=7.5m	0.000554	8.12	10.1	0.17	Yes	72.0
	EGO, horizon=10m	0.000844	8.17	10.3	0.31	Yes	94.1
	Proposed, $\kappa = 5$	**<u>0.000465</u>**	7.57	**9.70**	0.09	Yes	30.4
	Proposed, $\kappa = 8$	0.000557	7.58	9.71	0.11	Yes	**28.2**

Except for **EGO** (*33*) and **MADER** (*37*), we compare the proposed planner against two other centralized offline swarm planners, i.e., **SCP** (*49*) and **RBP** (*50*). Results are given in Table S1. A bold term indicates a better value in the corresponding category (Online/Offline), while an underlined term indicates the best value among all planners. As given in Table S1, **SCP** achieves a shorter flight time at the sacrifice of trajectory safety. **RBP** achieves smoother flight at the cost of a relatively long flight time. Both fail to balance various aspects of trajectory quality. Compared with offline methods, the three online planners show more balance while requiring less computation by several orders of magnitude. Among them, **MADER** is more conservative because its safety is ensured through minimum volume enclosing simplices. All the data are recorded from 10 random runs.

第S1节 互惠避碰的额外比较

表S1. 额外性能比较。 在一个包含八个半径为0.25米的智能体的 8×8 米空旷空间中，采用位置交换配置进行比较。最大速度和加速度分别设定为1.7米/秒和6米/秒^2, 。**最小距离。** 表示与碰撞的最小距离。**平滑度** 通过平方加加速度的时间积分计算得出 $\int ||j||_2^2 dt$。时间和距离的单位分别为秒和米。

线上/ 离线	方法	比较 时间	轨迹 时间	轨迹 长度	最小值 清除。	安全性	平滑- 性
离线	SCP，h_scp=0.3s	1.46	7.65	**<u>9.64</u>**	-0.37	无	7.01
	SCP，h_scp=0.17s	7.72	**<u>7.40</u>**	9.70	-0.26	否	17.5
	RBP 没有批次	**0.320**	15.4	11.3	0.01	是	0.608
	RBP 批量大小=4	0.413	15.4	11.5	0.03	是的	**<u>0.604</u>**
在线	Mader’	0.0239	**7.50**	12.2	0.19	是	125.1
	自我 horizon=7.5m	0.000554	8.12	10.1	0.17	是	72.0
	自我’地平线=10米	0.000844	8.17	10.3	0.31	是	94.1
	提出的，$\kappa = 5$	**<u>0.000465</u>**	7.57	**9.70**	0.09	是	30.4
	提出的，$\kappa = 8$	0.000557	7.58	9.71	0.11	是	**28.2**

除 **EGO** (*33*) 和 **MADER** (*37*) 外，我们将所提出的规划器与另外两种集中式离线群体规划器进行比较，即 **SCP** (*49*) 和 **RBP** (*50*)。结果见表 S1。加粗的术语表示在相应类别（在线/离线）中具有更优的数值，而带下划线的术语表示在所有规划器中最佳的数值。如表 S1 所示，**SCP** 在牺牲轨迹安全的情况下实现了更短的飞行时间。**RBP** 在飞行时间相对较长的代价下实现了更平滑的飞行。两者都未能在轨迹质量的各个方面取得平衡。与离线方法相比，这三种在线规划器表现出更好的平衡性，同时所需计算量少了几个数量级。其中，**MADER** 更加保守，因为其安全性通过最小体积包络单纯形得到保证。所有数据均来自10次随机运行的记录。

Section S2. Scalability Evaluation

Time and Space Complexity

In real-world usage, larger swarms always mean a more powerful team for broader coverage in exploration, a grater possible load for transportation, and higher robustness against robot failures, among other advantages. However, traditional optimization-based trajectory planning suffers from a high computing complexity related to the number of robots, as stated by Park et al. (*50*) the time complexity of which is $O(n^3)$. Therefore, Park et al. have to divide the entire swarm into groups and then plan the trajectory sequentially, which only reliefs the complexity to a limited extent.

Owing to the decentralized architecture and appropriate problem formulation, our time complexity is significantly reduced to approximately linear to the drone number from statistics. Since the baseline of computing time is short, the swarm size can increase to hundreds of robots before breaking the real-time bottom line. Meanwhile, the space complexity is exactly linear to the drone number, as one drone only stores the latest trajectory from every other one. Indeed, this space usage can be ignored because one trajectory is typically below 1 kB in size, and there are several gigabytes in the system memory.

The complexity can be further reduced (*20*) if each drone only considers nearby units, since we perform continuous planning only within a limited distance. In our work, we ignore drones more than twice the planning distance (see Table S6) away from each other. The resulting complexity is below $O(n)$, and is determined by the density of drone distribution as in Fig. S1. Such complexity allows the swarm to be expanded to a large size before breaking the real-time threshold.

第S2节 规模扩展性评估

时间与空间复杂度

在实际应用中，较大的蜂群总是意味着更强大的团队，以实现更广泛的探索覆盖、更大的运输负载潜力，以及对机器人故障的更高鲁棒性等优势。然而，基于传统优化的轨迹规划在计算复杂度方面受到机器人数量的显著影响，正如Park等人所指出（*50*），其时间复杂度为 $O(n^3)$。因此，Park等人不得不将整个蜂群划分为多个组，然后依次规划轨迹，这仅在一定程度上缓解了复杂度。

由于采用去中心化架构和合理的问题建模，我们的时间复杂度显著降低，统计数据显示其大致线性于无人机数量。由于计算时间的基线较短，蜂群规模可以增加到数百台机器人而不突破实时性底线。同时，空间复杂度恰好与无人机数量呈线性关系，因为每台无人机只存储来自其他所有无人机的最新轨迹。实际上，这种空间使用可以忽略不计，因为一条轨迹的大小通常低于1 kB，而系统内存中有数GB的空间。

如果每个无人机只考虑附近的单元，复杂度可以进一步降低（*20*），因为我们只在有限距离内进行连续规划。在我们的工作中，我们忽略了彼此距离超过两倍规划距离的无人机（见表 S6）。由此产生的复杂度低于 O（n），并且由图 S1 中的无人机分布密度决定。这种复杂度使得无人机群可以在突破实时阈值之前扩展到较大的规模。

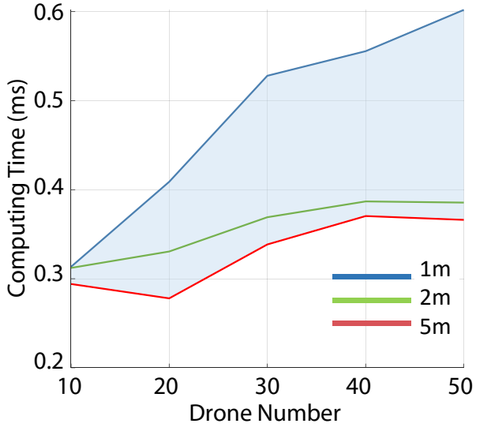
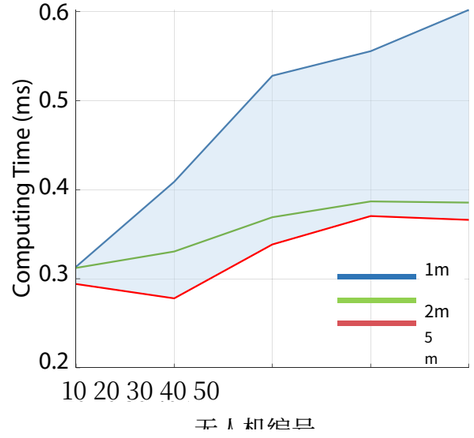


Fig. S1. Scalability on computation efficiency. This figure provides the computing time for different drone numbers under three swarm distributions shown in Fig. S15.

Wireless Communication Quality

Communication capacity can be affected by environmental signal interference, connection topology, and drone distribution. To control signal interference, we conduct this experiment in a quiet park with drones hovering and UWBs turned on. Then, in this test, we compare two topologies, namely a centralized network with one access point and a peer-to-peer ad hoc network using the hardware described in Section *Palm-Sized Drone Hardware*. In addition, various distributions illustrated in Fig. S2A from tight to loose, are tested. Two widely used metrics, bandwidth, and latency, are adopted to assess network capacity. Since we write reliable network communication software that guarantees data arrival, the packet loss rate is not evaluated (which equals either 0% or 100%). The bandwidth B_w is quantized to the highest trajectory (175 bytes each) broadcasting frequency by all the drones together before reaching the 250-ms latency threshold. Latency T_L is recorded at 5 Hz of the trajectory broadcasting rate.

To be collision-free, in a centralized network, each drone must maintain a direct connection to the access point (AP), so we record the worst value among all drones. By contrast, for the



图S1. 计算效率的可扩展性。本图展示了在图S15所示的三种蜂群分布下，不同无人机数量的计算时间。

无线通信质量

通信能力可能受到环境信号干扰、连接拓扑结构和无人机分布的影响。为了控制信号干扰，我们在一个安静的公园中进行此实验，无人机悬停且超宽带（UWB）已开启。在此测试中，我们比较了两种拓扑结构，即具有一个接入点的集中式网络和使用第 “掌上型无人机硬件” 中描述的硬件的点对点自组网。此外，还测试了图 S2A 中从紧密到松散的各种分布。采用两个广泛使用的指标——带宽和延迟，来评估网络容量。由于我们编写了可靠的网络通信软件，保证数据到达，因此不评估包丢失率（包丢失率要么为 0%，要么为 100%）。带宽 B_w 被量化为所有无人机共同广播的最高轨迹（每个175字节）频率，直到达到 250 毫秒的延迟阈值。延迟 T_L 在轨迹广播速率为 5 Hz 时记录。

为了避免碰撞，在集中式网络中，每个无人机必须与接入点（AP）保持直接连接，因此我们记录所有无人机中最差的值。相比之下，对于

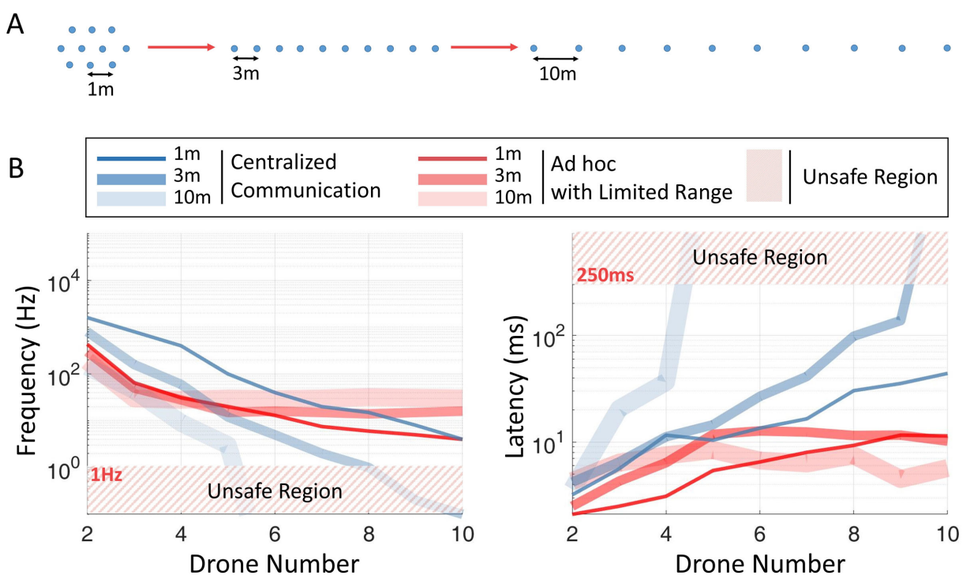
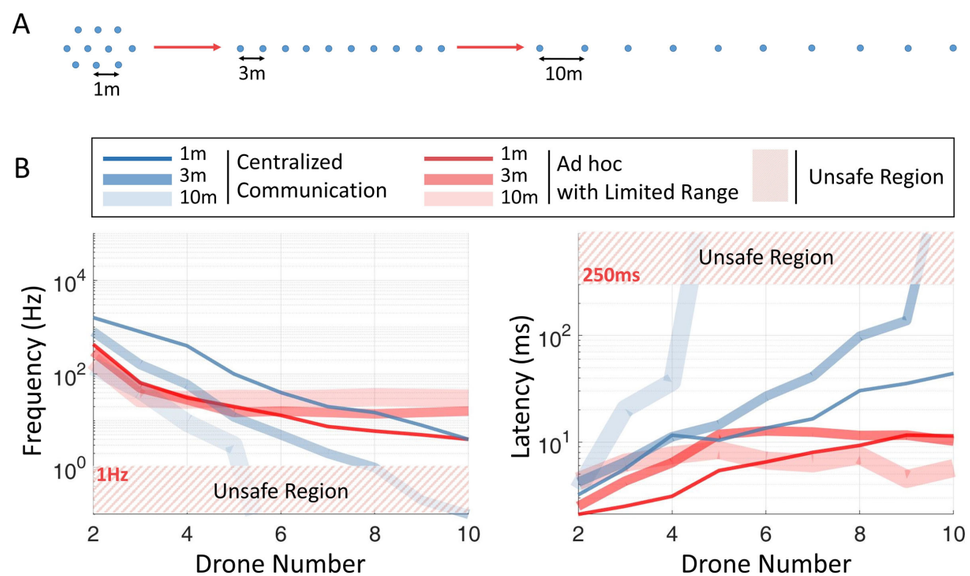


Fig. S2. Scalability on communication quality. (**A**) Three kinds of swarm distributions. (**B**) Network quality for two kind of topologies under three swarm distributions.

ad hoc network, each drone directly connects to the others. Furthermore, when two drones get further than twice the planning horizon from each other, the collision penalty between them is always zero, and freedom from collision is guaranteed naturally. Thus, we only evaluate data between each pair of drones with a relative distance below twice the planning horizon (15 m in this experiment) for an ad hoc network.

From the results in Fig. S2B, a centralized network performs better in tight formation, while an ad hoc network shows better communication quality when the formation is loose. From the integral view, an ad hoc network scales better with the number of drones when in loose distribution, while a centralized network provides higher bandwidth when the drone number is small; thus, both networks are complementary to an extent. In theory, as a peer-to-peer network, an ad hoc network creates connections that follow factorial complexity in a dense formation, while a



图S2. 通信质量的可扩展性。（**A**）三种群分布方式。（**B**）在三种群分布下两种拓扑结构的网络质量。

自组网，每个无人机直接连接到其他无人机。此外，当两个无人机之间的距离超过规划视野的两倍（在本实验中为15米）时，它们之间的碰撞惩罚始终为零，自然保证了无碰撞自由。因此，我们仅对相对距离低于两倍规划视野（15米，在本实验中）的每对无人机之间的数据进行评估，适用于自组网。

从图S2B中的结果可以看出，集中式网络在紧密编队中表现更佳，而在松散编队时，临时网络显示出更好的通信质量。从整体来看，临时网络在松散分布情况下随着无人机数量的增加具有更好的扩展性，而集中式网络在无人机数量较少时提供更高的带宽；因此，这两种网络在一定程度上是互补的。从理论上讲，作为一种点对点网络，临时网络在密集编队中创建的连接遵循阶乘复杂度，而集中式网络在这种场景下仅遵循线性复杂度。因此，开发者必须根据任务特性选择合适的网络拓扑结构，或采用两者的结合。所提出的无人机平台支持这两种网络。

centralized network only follows linear complexity in this scenario. Therefore, developers must choose a proper network topology or adopt both types, of them according to task characteristics. The proposed drone platform supports both networks.

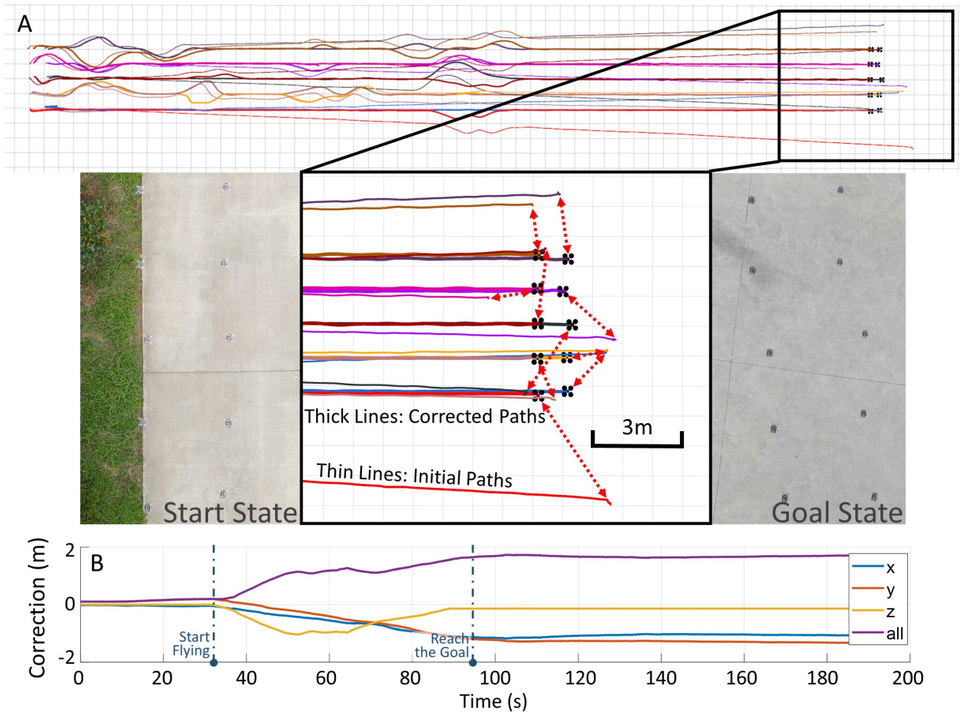


Fig. S3. Localization drift corrections. (**A**) Thin and thick lines are trajectories before and after corrections, respectively. The former ones show significant drifts that could have ruined the formation if not corrected. (**B**) Localization drift of drone 0. More drones are shown in Fig. S13.

Section S3. Assessments of Localization Drift Correction

The data recorded from the experiment in Section *Formation Navigation in the Wild* are given in Fig. S3. The desired start and goal shapes are 2×5 drone rectangles with 1-m spacings. With

集中式网络在此场景下仅表现出线性复杂度。因此，开发者必须选择合适的网络拓扑结构，或根据任务特性同时采用两者。所提出的无人机平台支持这两种网络。

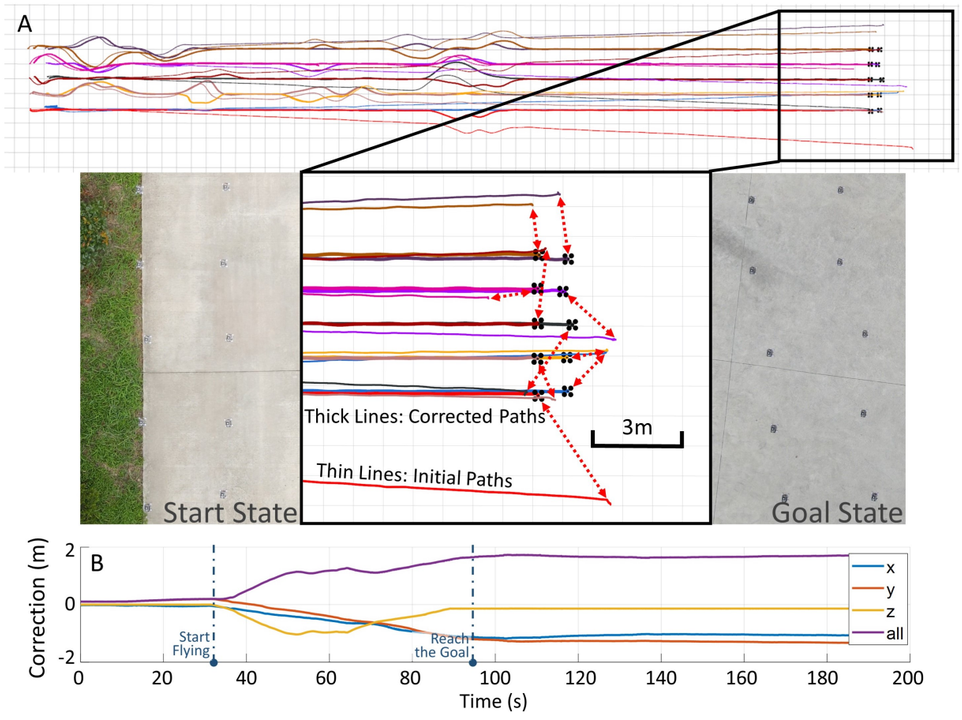


图 S3. 定位漂移校正。(**A**) 细线和粗线分别表示校正前后的轨迹。前者显示了可能会破坏队形的显著漂移，如果不进行校正的话。(**B**) 无人机 0 的定位漂移。更多无人机的情况见图 S13。

第 S3 节 定位漂移校正的评估

第野外编队导航部分实验中记录的数据如图S3所示。期望的起始和目标形状为 2×5 无人机矩形，间距为1米。通过

the corrected localization, the drone managed to maintain the shape after a 55-m flight, although they are not precisely aligned in yaw at the start point. By contrast, Figs. S3 and S13 show the magnitude of the estimated drift, which reaches approximately 3 m. Under such significant drift, the formation suffers severe shape distortion and reciprocal collision if no correction is made. However, since no global information such as GPS data is available, accumulated errors for the entire team must exist. That is why we add some fixed anchors in Section *Intensive Reciprocal Avoidance Evaluation*. In summary, the necessity and effectiveness of the proposed correction method are validated here. The block diagram is shown in Fig. S14.

Section S4. Quantitative Analysis of the Gate-passing Test

To further analyze the planner performance, the gate-passing test in Section *Benchmark Comparisons* is conducted again with some representative data visualized in Fig. S4. Among the four sub-figures, B and C are mutually constrained. This means, for a shorter flight time, these drones must fly as fast as possible under the maximum speed of 2 m/s, while they also have to alter velocity to maintain a relative distance. Then, for drones in front, velocity can be increased to reduce flight time while enlarging relative distance, but the velocity is limited. For the drones left behind, reducing the velocity as little as possible to maintain the shortest relative distance to those in front is the best strategy. The above is exactly what is achieved in Fig. S4B and C, where drones managed to fly across the gate at the maximum velocity and minimum relative distance allowed. What's more, from Fig. S4D, we can conclude that drones fly along the shortest trajectory without detours or wasting of energy.

Section S5. Trajectory Parameterization for Aerial Robots

The pros and cons of typical trajectory parameterization for aerial robots are summarized in Table S2. **Polynomial** curves are widely used by traditional approaches (*4, 10, 38*), which mostly

经过校正的定位后，无人机在55米飞行后仍能保持队形，尽管它们在起点的偏航角未能完全对齐。相比之下，图S3和S13显示了估计漂移的幅度，达到大约3米。在如此显著的漂移情况下，如果不进行校正，队形会受到严重的形状扭曲和相互碰撞。然而，由于没有全球信息如GPS数据，整个队伍的累积误差必然存在。这也是我们在密集互避评估部分添加一些固定锚点的原因。总之，本文验证了所提出的校正方法的必要性和有效性。框图如图S14所示。

节 n S4. 门控测试的定量分析

为了进一步分析规划器的性能，重新进行了第 基准比较节中的门控测试，并用一些具有代表性的数据在图 S4 中进行了可视化。在这四个子图中，B 和 C 彼此受限。这意味着，为了实现更短的飞行时间，这些无人机必须在最大速度 2 m/s 下尽可能快地飞行，同时还必须改变速度以保持相对距离。然后，对于前方的无人机，可以通过增加速度来缩短飞行时间，同时扩大相对距离，但速度受到限制。对于落后方的无人机，保持最短的相对距离以尽可能少地降低速度是最佳策略。上述情况正是在图 S4B 和 C 中实现的，无人机成功以最大速度和允许的最小相对距离飞越了门控。此外，从图 S4D 可以得出结论，无人机沿着最短轨迹飞行，没有绕道或能量浪费。

第 S5 节 空中机器人轨迹参数化

典型轨迹参数化方法的优缺点总结见表 S2。**多项式**曲线被传统方法广泛采用（*4,10,38*），这些方法大多

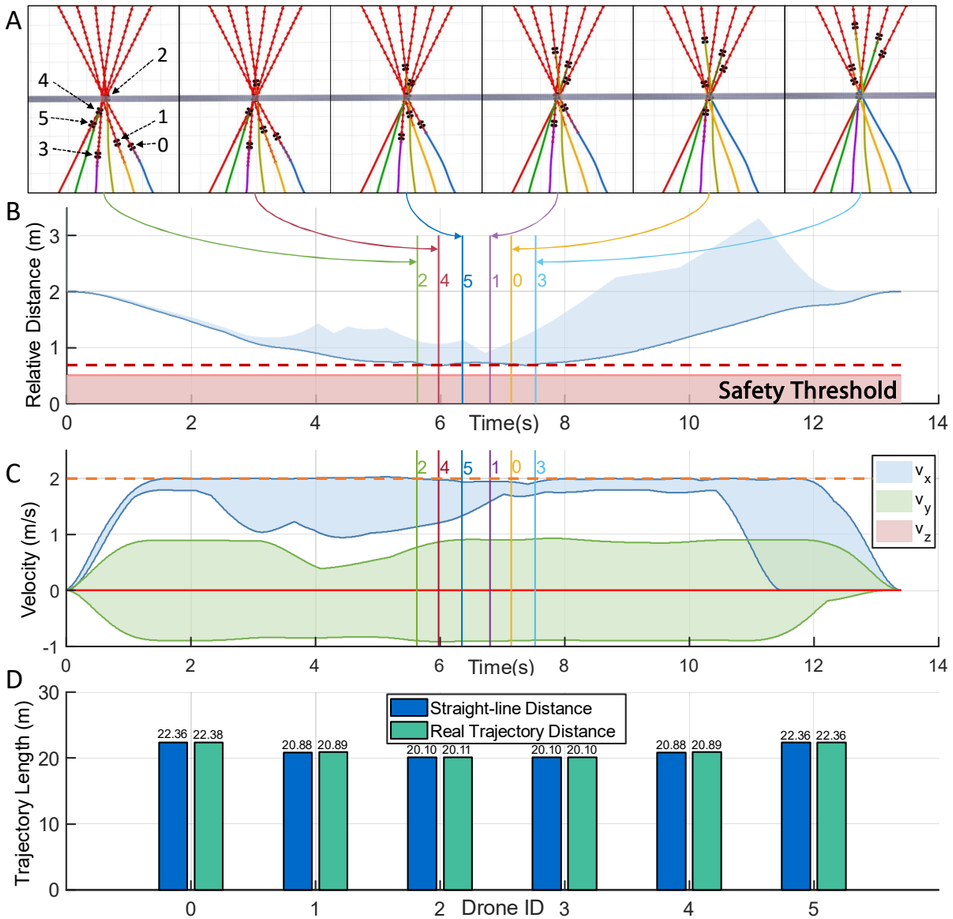


Fig. S4. Performance evaluation in the gate-passing test. (**A**) Six snapshots taken when each drone is passing the gate. (**B**) The minimum relative distances recorded by each drone are denoted by the light blue area, the lower bound of which is emphasized by a blue curve. Multicolored lines marked with numbers of drone ID indicate the time at which each drone flies through the gate. Red dotted line is the 0.7-m swarm clearance in optimization. (**C**) Minimum-to-maximum velocity curves. The orange dotted line at a 2-m/s velocity is the velocity limit. (**D**) Comparison between real trajectory lengths and straight-line distances (the minimum length in theory).

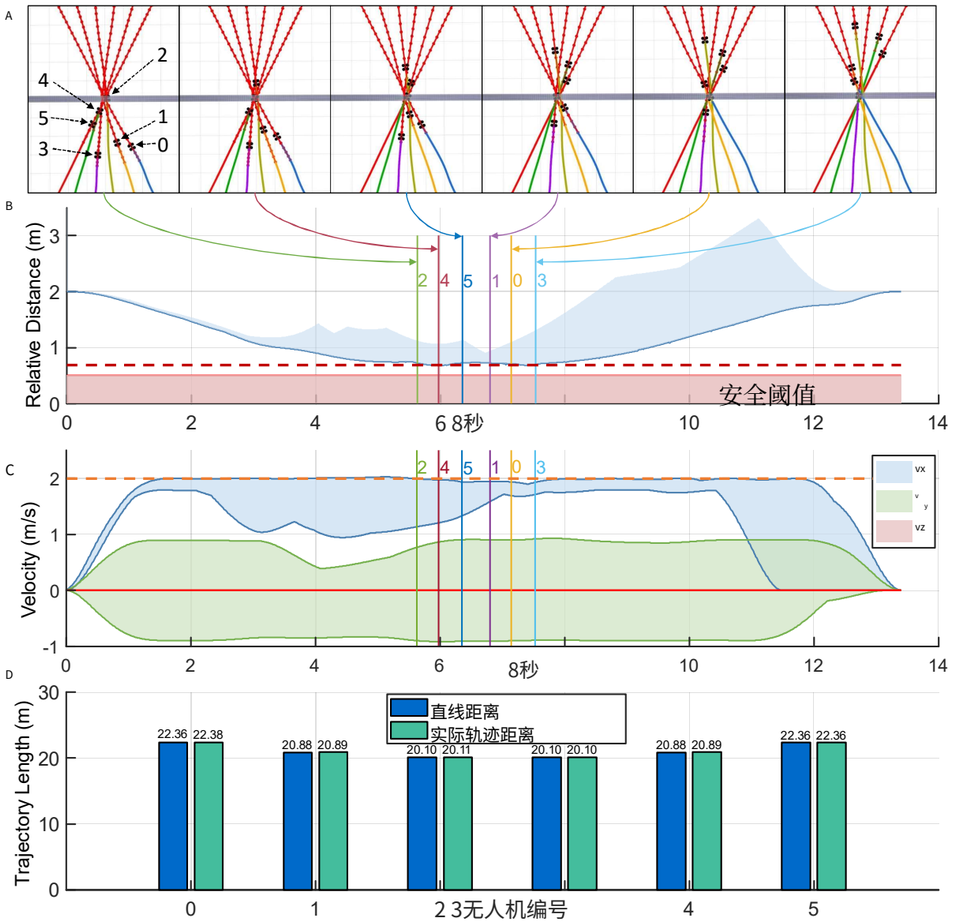


图 S4. 通行门测试中的性能评估。 (**A**) 每架无人机通过门时拍摄的六个快照。 (**B**) 每架无人机记录的最小相对距离由浅蓝色区域表示，其下限由蓝色曲线强调。标有无人机ID的多彩线条表示每架无人机飞过门的时间。红色虚线是在优化中设定的0.7米的群体间距。 (**C**) 最小到最大速度曲线。以2米/秒速度的橙色虚线为速度限制。 (**D**) 实际轨迹长度与直线距离（理论上的最小长度）之间的比较。

use polynomial coefficients as decision variables, along with explicit continuity constraints. Many works (*10, 38*) deform a piece-wise polynomial trajectory by adding intermediate waypoints. However, the distribution of waypoints requires a careful configuration to balance feasibility satisfaction and computational burden. To optimize the time profile of a polynomial, the representative method uses gradient descent with finite differences (*38*). Nevertheless, this method requires a dense matrix inversion to transform waypoints to coefficients, making the optimization quickly intractable when the problem scales up. **Bézier** and **B-Spline** share the *convex hull* nature, which indicates that the curve is entirely contained in the convex hull of its control points (*13, 23*). Therefore, these curves are convenient for adding constraints, which can be done by merely restricting the control points inside feasible convex regions. Many works (*24, 37, 50*) follow this underlying property and show successful applications in aerial swarms. However, this property indeed brings conservativeness, as analyzed by (*47*), preventing a trajectory from being aggressive near its physical limits. Moreover, an n order **B-Spline** is naturally $n-1$ order continuous (*48*). Thus no continuity constraint is required as long as the B-Spline has a qualified order. The time profile of a **Bézier** curve can be adjusted with acceptable complexity by its basis (*23*), while this approach is rather complicated for a B-Spline due to its time evaluation containing high nonlinearity. **MINCO** (*57*) is a newly developed trajectory representation specialized for integrator chain systems, and its core feature is that it efficiently handles a wide variety of constraints while retaining spatial and temporal optimality. By discretizing the trajectory to add constraints, MINCO is less conservative but requires further checks after obtaining the solution. Although MINCO is the most complicated representation and requires a sophisticated implementation, it provides a solid foundation for spatial-temporal drone swarm planning.

将多项式系数作为决策变量，并结合显式连续性约束。许多工作（*10,38*）通过添加中间路径点来变形分段多项式轨迹。然而，路径点的分布需要仔细配置，以在可行性满足和计算负担之间取得平衡。为了优化多项式的时间轮廓，代表性方法采用有限差分的梯度下降（*38*）。然而，该方法需要进行密矩阵求逆，将路径点转换为系数，当问题规模扩大时，优化变得迅速难以处理。**B´贝塞尔**和**B样条**共享凸包特性，这表明曲线完全包含在其控制点的凸包内（*13,23*）。因此，这些曲线在添加约束时非常方便，只需限制控制点在可行的凸区域内即可。许多工作（*24, 37,50*）遵循这一基本性质，并在空中群体中展示了成功的应用。然而，这一性质确实带来了保守性，正如（*47*）所分析的那样，阻止轨迹在其物理极限附近变得激进。此外，n 阶**B样条**本身具有 $n-1$ 阶连续性（*48*）。因此，只要B样条具有合格的阶数，就不需要连续性约束。**B´贝塞尔**曲线的时间轮廓可以通过其基函数（*23*）以可接受的复杂度进行调整，而对于B样条，由于其时间评估包含高度非线性，这种方法相当复杂。**MINCO**（*57*）是一种新开发的轨迹表示方法，专为积分链系统设计，其核心特性是高效处理各种约束，同时保持空间和时间的最优性。通过离散化轨迹以添加约束，MINCO的保守性较低，但在获得解后需要进行进一步检查。尽管MINCO是最复杂的表示方式，且需要精细的实现，但它为空间-时间无人机群规划提供了坚实的基础。

Table S2. Pros and cons of commonly used trajectory parameterizations.

Method	Continuity	Safety	Dynamical Feasibility	Temporal Optimization	Implementation Difficulty
Polynomial	Extra Overhead	By Discretization		Coupled	Easy
Bézier Curve		Analytical but Conservative			Medium
B-Spline	By Nature			Intractable	
MINCO		By Discretization		Decoupled	

Section S6. Spatial-Temporal Trajectory Optimization for Swarms — A More General and Detailed Problem Formulation

MINCO Trajectory Class

MINCO is a polynomial trajectory class defined as

$$\text{MINCO} = \Big\{ p(t) : [0, t_M] \mapsto \mathbb{R}^m \Big| \ \mathbf{c} = \mathcal{M}(\mathbf{q}, \mathbf{T}), \\ \mathbf{q} \in \mathbb{R}^{m \times M-1}, \ \mathbf{T} \in \mathbb{R}^M_{>0} \Big\},$$

where $p(t)$ is an m-dimensional M-piece polynomial trajectory with degree $N, N = 2s-1$ with s the order of the relevant integrator chain, $\mathbf{c} = (\mathbf{c}_1^{\mathrm{T}}, \ldots, \mathbf{c}_M^{\mathrm{T}})^{\mathrm{T}} \in \mathbb{R}^{2Ms \times m}$ the polynomial coefficient with $\mathbf{c}_i \in \mathbb{R}^{2s \times m}$ the coefficient matrix, $\mathbf{q} = (\mathbf{q}_1, \ldots, \mathbf{q}_{M-1})$ the intermediate waypoints, $\mathbf{T} = (T_1, T_2, \ldots, T_M)^{\mathrm{T}}$ the time allocated for all pieces, $t_i = \sum_{j=1}^{i} T_j, \ i \in [1, M]$, and $t_0 = 0$ in the Supplementary Materials. The notion $\mathcal{M}(\mathbf{q}, \mathbf{T})$ equals $\mathcal{M}_{\mathbf{q},\mathbf{T}}$ in the main text. Specifically, the trajectory is evaluated as

$$p(t) = p_i(t - t_{i-1}), \ \forall t \in [t_{i-1}, t_i]. \tag{S1}$$

The i-th piece $p_i(t) : [0, T_i] \mapsto \mathbb{R}^m$ is defined by

$$p_i(t) = \mathbf{c}_i^{\mathrm{T}} \beta(t), \ \forall t \in [0, T_i], \tag{S2}$$

表S2. 常用轨迹参数化方法的优缺点。

方法	连续性	安全性	动力学可行性	时间优化	实现难点
多项式	额外俯视图	通过离散化		耦合	简单
B´贝塞尔曲线		解析但保守			中等
B样条	本质上			难以处理	
MINCO		离散化		解耦	

第S6节。群体的时空轨迹优化——一个更一般且详细的问题公式

MINCO轨迹类别

MINCO 是一种多项式轨迹类别，定义为

$$\text{MINCO} = \Big\{ p(t) : [0, t_M] \mapsto \mathbb{R}^m \Big| \ \mathbf{c} = \mathcal{M}(\mathbf{q}, \mathbf{T}), \\ \mathbf{q} \in \mathbb{R}^{m \times M-1}, \ \mathbf{T} \in \mathbb{R}^M_{>0} \Big\},$$

其中 $p(t)$ 是一个 m 维 M 段多项式轨迹，阶数为 N，$N = 2s-1$，其中 s 表示相关积分链的阶数，$\mathrm{c} = (\mathrm{c}_1^{\mathrm{T}}, \ldots, \mathrm{c}_M^{\mathrm{T}})^{\mathrm{T}} \in \mathbb{R}^{2Ms \times m}$ 为多项式系数，$\mathrm{c}_i \in \mathbb{R}^{2s \times m}$ 为系数矩阵，$\mathrm{q} = (\mathrm{q}1, \ldots, \mathrm{q}_{M-1})$ 为中间路径点，$\mathrm{T} = (T_1, T_2, \ldots, T_M)^{\mathrm{T}}$ 表示所有段的时间分配，$t_i = \sum_{j=1}^{i} T_j, \ i \in [1, M]$，以及$t_0 = 0$ 在补充材料中。符号 $\mathcal{M}(\mathrm{q}, \mathrm{T})$ 等同于主文中的 $\mathcal{M}_{\mathbf{q},\mathbf{T}}$。具体而言，轨迹的评估如下：

$$p(t) = p_i(t - t_{i-1}), \ \forall t \in [t_{i-1}, t_i]. \tag{S1}$$

第 i 片段 $p_i(t)$：$[0, T_i] \mapsto \mathbb{R}^m$ 由以下定义

$$p_i(t) = \mathbf{c}_i^{\mathrm{T}} \beta(t), \ \forall t \in [0, T_i], \tag{S2}$$

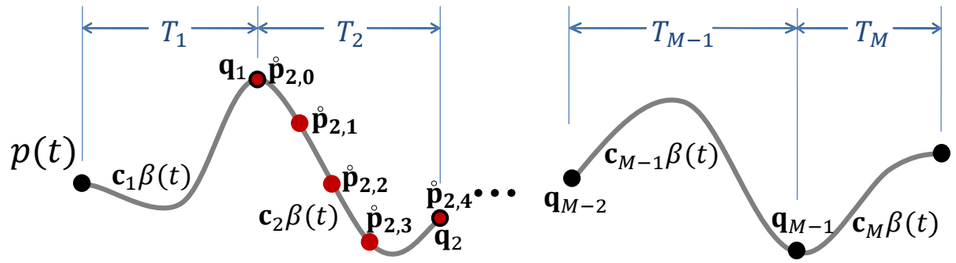


Fig. S5. Parameters of a MINCO trajectory and its constraint evaluation. The trajectory is directly parameterized by each waypoint $\mathbf{q}_i$ and time T_i. Red dots are *constraint points* $\mathring{\mathbf{p}}$ at which any constraint is evaluated and propagated to every $\mathbf{q}_i$ and T_i.

where $\beta(t) := [1, t, \cdots, t^N]^{\mathrm{T}}$ is the natural basis. The core of MINCO is the parameter mapping $\mathbf{c} = \mathcal{M}(\mathbf{q}, \mathbf{T})$ constructed from Theorem 2 in (*57*). Technically, the mapping directly constructs a minimum control trajectory for an m-dimensional s-integrator chain for any specified initial and terminal conditions. An instance in MINCO is intuitively shown in Fig. S5. Owing to the limited paper length, we only intuitively introduce the features of MINCO. Detailed proofs are presented in (*57*).

Feature 1: Compactly represented by $\mathbf{q}$ and $\mathbf{T}$, MINCO is a class of $\mathcal{C}^{s-1}$ polynomial trajectories satisfying the following control effort minimization:

$$\min_{p(t)} \int_{t_0}^{t_M} p^{(s)}(t)^{\mathrm{T}} \mathbf{W} p^{(s)}(t) \mathrm{d}t, \tag{S3a}$$

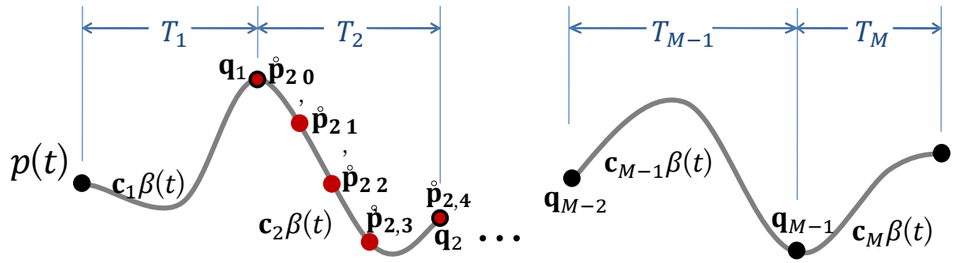
$$s.t.\ \ p^{[s-1]}(t_0) = \bar{p}_o,\ p^{[s-1]}(t_M) = \bar{p}_f, \tag{S3b}$$

$$p(t_i) = \mathbf{q}_i,\ 1 \le i < M, \tag{S3c}$$

$$t_{i-1} < t_i,\ 1 \le i \le M, \tag{S3d}$$

where $\mathbf{W} \in \mathbb{R}^{m \times m}$ is a diagonal matrix with positive entries, $p^{[s-1]} = [p, p^{(1)}, p^{(2)}, \cdots, p^{(s-1)}]$ with , $\bar{p}_o$ and $\bar{p}_f \in \mathbb{R}^{m \times s}$ the specified initial and terminal conditions, respectively. The trajectory is enforced to pass $\mathbf{q}_i$ at time t_i.

Feature 2: The evaluation and differentiation of the mapping $\mathbf{c} = \mathcal{M}(\mathbf{q}, \mathbf{T})$ benefits from



图S5. MINCO轨迹的参数及其约束评估。轨迹由每个路径点 $\mathbf{q}_i$ 和时间 T_i 直接参数化。红点为约束点 $\mathring{\mathbf{p}}$，在这些点上对任何约束进行评估并传播到每个$\mathbf{q}_i$ 和 T_i。

其中 $\beta(t) := [1, t, \cdots, t^N]^{\mathrm{T}}$ 是自然基。MINCO 的核心是参数映射 $\mathbf{c} = \mathcal{M}(\mathbf{q},\mathbf{T})$，该映射由 (*57*) 中的定理 2 构建。从技术上讲，该映射直接为任何指定的初始和终止条件构建一个最小控制轨迹，用于一个 m 维的 s 积分链。MINCO 中的一个实例直观地显示在图 S5 中。由于论文篇幅有限，我们仅直观介绍 MINCO 的特性。详细的证明在 (*57*) 中给出。

特征1： 由 $\mathbf{q}$ 和 $\mathbf{T}$紧凑表示，MINCO是一类满足以下控制努力最小化的 $\mathcal{C}^{s-1}$ 多项式轨迹：

$$\min_{p(t)} \int_{t_0}^{t_M} p^{(s)}(t)^{\mathrm{T}} \mathbf{W} p^{(s)}(t) \mathrm{d}t, \tag{S3a}$$

$$s.t.\ \ p^{[s-1]}(t_0) = \bar{p}_o,\ p^{[s-1]}(t_M) = \bar{p}_f, \tag{S3b}$$

$$p(t_i) = \mathbf{q}_i,\ 1 \le i < M, \tag{S3c}$$

$$t_{i-1} < t_i,\ 1 \le i \le M, \tag{S3d}$$

其中 $\mathbf{W} \in \mathbb{R}^{m \times m}$ 是一个对角矩阵，具有正元素，$p^{[s-1]} = [p, p^{(1)}, p^{(2)}, \cdots, p^{(s-1)}]$ 表示，$\bar{p}_o$ 和 $\bar{p}_f \in \mathbb{R}^{m \times s}$ 分别为指定的初始和终止条件。轨迹被强制通过时间 t_i 处的 $\mathbf{q}_i$。

特征 2： 映射 $\mathbf{c} = \mathcal{M}(\mathbf{q},\mathbf{T})$ 的评估与区分依赖于线性复杂度。更具体的对应关系可以用以下函数表示

linear complexity. A more specific correspondence can be expressed as the following function

$$\mathbf{M}(\mathbf{T})\mathbf{c} = \mathbf{b}(\mathbf{q}), \tag{S4}$$

where $\mathbf{M}(\mathbf{T}) \in \mathbb{R}^{2Ms\times 2Ms}$ is a banded matrix with non-singularity for any $\mathbf{T} \succ \mathbf{0}$ ensured by Theorem 2 in (*57*), $\mathbf{b}(\mathbf{q}) \in \mathbb{R}^{2Ms\times m}$. The conversion between these two trajectory representations, i.e., $(\mathbf{c}, \mathbf{T})$ and $(\mathbf{q}, \mathbf{T})$ is achieved using *Banded PLU Factorization* with $O(M)$ linear time and space complexity. The evaluation is extremely fast, which takes approximately 1 μs per piece for minimum-jerk ($s = 3$) trajectory generation on a desktop CPU.

The formulation of $\mathbf{M}$ and $\mathbf{b}$ are

$$\mathbf{M} = \begin{pmatrix} \mathbf{F}_0 & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{E}_1 & \mathbf{F}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{E}_2 & \mathbf{F}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{F}_{M-1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{E}_M \end{pmatrix}, \tag{S5a}$$

$$\mathbf{E}_i = (\beta^{(0)}(T_i), \beta^{(0)}(T_i), \beta^{(1)}(T_i), \cdots, \beta^{(2s-2)}(T_i))^{\mathrm{T}} \in \mathbb{R}^{2s\times 2s},\ i \in [1, M-1] \tag{S5b}$$

$$\mathbf{F}_i = (\mathbf{0}_{2s\times 1}, -\beta^{(0)}(0), \cdots, -\beta^{(2s-2)}(0))^{\mathrm{T}} \in \mathbb{R}^{2s\times 2s},\ i \in [1, M-1] \tag{S5c}$$

$$\mathbf{E}_M = \left(\beta^{(0)}(T_M), \cdots, \beta^{(s-1)}(T_M)\right)^{\mathrm{T}} \in \mathbb{R}^{s\times 2s}, \tag{S5d}$$

$$\mathbf{F}_0 = \left(\beta^{(0)}(0), \cdots, \beta^{(s-1)}(0)\right)^{\mathrm{T}} \in \mathbb{R}^{s\times 2s}; \tag{S5e}$$

$$\mathbf{b} = \begin{pmatrix} \mathbf{D}_0, & \mathbf{D}_1, & \mathbf{0}_{m\times(2s-1)}, & \cdots, & \mathbf{D}_{M-1}, & \mathbf{0}_{m\times(2s-1)}, & \mathbf{D}_M \end{pmatrix}^{\mathrm{T}}, \tag{S6a}$$

$$\mathbf{D}_i = \mathbf{q}_i \in \mathbb{R}^m,\ i \in [1, M-1], \tag{S6b}$$

$$\mathbf{D}_0 = \bar{p}_o,\ \mathbf{D}_M = \bar{p}_f,\ \mathbf{D}_0, \mathbf{D}_M \in \mathbb{R}^{m\times s}. \tag{S6c}$$

Feature 3: Feature 2 allows any user-defined objective or constraint function $F(\mathbf{c}, \mathbf{T})$ with available gradients applicable to MINCO parameterized in $(\mathbf{q}, \mathbf{T})$. Specifically, the correspond-

线性复杂度。更具体的对应关系可以用以下函数表示

$$\mathbf{M}(\mathbf{T})\mathbf{c} = \mathbf{b}(\mathbf{q}), \tag{S4}$$

其中 $\mathbf{M}(\mathbf{T}) \in \mathbb{R}^{2Ms\times 2Ms}$ 是一个带状矩阵，对于任何 $\mathbf{T} \succ \mathbf{0}$，都具有非奇异性，这由定理 2 在 (*57*) 中保证，$\mathbf{b}(\mathbf{q}) \in \mathbb{R}^{2Ms\times m}$。这两种轨迹表示法，即 $(\mathbf{c}, \mathbf{T})$ 和 $(\mathbf{q}, \mathbf{T})$ 之间的转换，是通过 带状 *PLU* 分解 实现的，具有 $O(M)$ 线性时间和空间复杂度。该评估非常快速，在桌面 CPU 上进行最小冲击 ($s = 3$) 轨迹生成时，每段大约耗时 1 μs。

$\mathbf{M}$ 和 $\mathbf{b}$ 的公式为

$$\mathbf{M} = \begin{pmatrix} \mathbf{F}_0 & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{E}_1 & \mathbf{F}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{E}_2 & \mathbf{F}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{F}_{M-1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{E}_M \end{pmatrix}, \tag{S5a}$$

$$\mathbf{E}_i = (\beta^{(0)}(T_i), \beta^{(0)}(T_i), \beta^{(1)}(T_i), \cdots, \beta^{(2s-2)}(T_i))^{\mathrm{T}} \in \mathbb{R}^{2s\times 2s},\ i \in [1, M-1] \tag{S5b}$$

$$\mathbf{F}_i = (\mathbf{0}_{2s\times 1}, -\beta^{(0)}(0), \cdots, -\beta^{(2s-2)}(0))^{\mathrm{T}} \in \mathbb{R}^{2s\times 2s},\ i \in [1, M-1] \tag{S5c}$$

$$\mathbf{E}_M = \left(\beta^{(0)}(T_M), \cdots, \beta^{(s-1)}(T_M)\right)^{\mathrm{T}} \in \mathbb{R}^{s\times 2s}, \tag{S5d}$$

$$\mathbf{F}_0 = \left(\beta^{(0)}(0), \cdots, \beta^{(s-1)}(0)\right)^{\mathrm{T}} \in \mathbb{R}^{s\times 2s}; \tag{S5e}$$

$$\mathbf{b} = \begin{pmatrix} \mathbf{D}_0, & \mathbf{D}_1, & \mathbf{0}_{m\times(2s-1)}, & \cdots, & \mathbf{D}_{M-1}, & \mathbf{0}_{m\times(2s-1)}, & \mathbf{D}_M \end{pmatrix}^{\mathrm{T}}, \tag{S6a}$$

$$\mathbf{D}_i = \mathbf{q}_i \in \mathbb{R}^m,\ i \in [1, M-1], \tag{S6b}$$

$$\mathbf{D}_0 = \bar{p}_o,\ \mathbf{D}_M = \bar{p}_f,\ \mathbf{D}_0, \mathbf{D}_M \in \mathbb{R}^{m\times s}. \tag{S6c}$$

特性 3： 特性 2 允许任何用户定义的目标或约束函数 $F(\mathbf{c}, \mathbf{T})$，并且具有适用于 MINCO 参数化的梯度，参数为 $(\mathbf{q}, \mathbf{T})$。具体而言，对应的目标函数 MINCO 被计算为

ing objective of MINCO is computed as

$$H(\mathbf{q},\mathbf{T}) = F(\mathcal{M}(\mathbf{q},\mathbf{T}),\mathbf{T}). \tag{S7}$$

Then the mapping $\mathbf{c} = \mathcal{M}(\mathbf{q},\mathbf{T})$ gives a linear-complexity way of computing $\partial H/\partial\mathbf{q}$ and $\partial H/\partial\mathbf{T}$ from the corresponding $\partial F/\partial\mathbf{c}$ and $\partial F/\partial\mathbf{T}$, i.e.,

$$\begin{array}{ccccccc} \dfrac{\partial H}{\partial \mathbf{q}} & = & \dfrac{\partial F}{\partial \mathbf{c}} & \dfrac{\partial \mathbf{c}}{\partial \mathbf{q}}, & \dfrac{\partial H}{\partial \mathbf{T}} & = & \dfrac{\partial F}{\partial \mathbf{c}}\dfrac{\partial \mathbf{c}}{\partial \mathbf{T}} + \dfrac{\partial F}{\partial \mathbf{T}}, \\ \uparrow & & \uparrow & \uparrow & \uparrow & & \uparrow \end{array} \tag{S8}$$

where derivatives marked by $\uparrow$ are the gradients finally required by a numerical optimizer, $\uparrow$ are the elements calculated by user code, and $\uparrow$ are just computed using the linear-complexity mapping $\mathbf{c} = \mathcal{M}(\mathbf{q},\mathbf{T})$, i.e., Eqs. S4 –S6. Users need not consider the constraints between $\mathbf{c}$ and $\mathbf{T}$ as if decoupled. Next, a high-level optimizer is able to optimize the objective efficiently.

Time Integral Constraints

Conventionally, requirements for dynamical feasibility and collision avoidance are formulated as functional-type constraints $\mathcal{G}(p(t),\dots,p^{(s)}(t)) \preceq \mathbf{0}, \forall t \in [0,t_M]$, $\mathcal{G} \in \mathbb{R}^{n_g}$ with n_g the number of types of constraints. However, this formulation cannot be directly handled by constrained optimization, since $\mathcal{G}$ contains infinitely many inequality constraints. Thus, we transform $\mathcal{G}$ into a finite-dimensional one via an integral of constraint violation. Practically, the integral is transformed into a weighted sum $J_\Sigma(\mathbf{c},\mathbf{T})$ of the sampled penalty function. In most cases where constraints are decoupled, i.e., constraints $\mathcal{G}(p^{[s]}(t))$ with $t_i \le t < t_{i+1}$ are merely determined by $\mathbf{c}_i$ and T_i, $p^{[s]} = [p, p^{(1)}, \cdots, p^{(s)}]$, the penalty for an i-th trajectory piece is computed as

$$J_i(\mathbf{c}_i,T_i) = \frac{T_i}{\kappa_i}\sum_{j=0}^{\kappa_i} \bar{\omega}_j \chi^{\mathrm{T}} \max\left(\mathcal{G}(\mathbf{c}_i,T_i,\frac{j}{\kappa_i}),\mathbf{0}\right)^3, \tag{S9}$$

where κ_i is the sample number on the i-th piece, $\chi \in \mathbb{R}^{n_g}_{\ge 0}$ a vector of penalty weights with appropriately large entries (several orders of magnitude larger than the smoothness penalty),

目标函数的计算公式为

$$H(\mathbf{q},\mathbf{T}) = F(\mathcal{M}(\mathbf{q},\mathbf{T}),\mathbf{T}). \tag{S7}$$

然后映射 $\mathbf{c} = \mathcal{M}(\mathbf{q},\mathbf{T})$ 提供了一种线性复杂度的计算 $\partial H/\partial\mathbf{q}$ 和$\partial H/\partial\mathbf{T}$ 的方法，基于相应的 $\partial F/\partial\mathbf{c}$ 和 $\partial F/\partial\mathbf{T}$，即，

$$\begin{array}{ccccccc} \dfrac{\partial H}{\partial \mathbf{q}} & = & \dfrac{\partial F}{\partial \mathbf{c}} & \dfrac{\partial \mathbf{c}}{\partial \mathbf{q}}, & \dfrac{\partial H}{\partial \mathbf{T}} & = & \dfrac{\partial F}{\partial \mathbf{c}}\dfrac{\partial \mathbf{c}}{\partial \mathbf{T}} + \dfrac{\partial F}{\partial \mathbf{T}}, \\ \uparrow & & \uparrow & \uparrow & \uparrow & & \uparrow \end{array} \tag{S8}$$

其中由 $\uparrow$ 标记的导数是数值优化器最终所需的梯度， $\uparrow$是由用户代码计算的元素，而 $\uparrow$ 仅通过线性复杂度映射 $\mathbf{c} = \mathcal{M}(\mathbf{q},\mathbf{T})$ 计算，即方程 S4 – S6。用户无需考虑 $\mathbf{c}$ 和 $\mathbf{T}$ 之间的约束，仿佛它们是解耦的。接下来，高层次的优化器能够高效地优化目标函数。

时间积分约束

传统上，关于动力学可行性和碰撞避免的要求被表述为函数型约束 $\mathcal{G}(p(t)\dots,p^{(s)}(t)) \preceq \mathbf{0}, \forall t \in [0,t_M]$, $\mathcal{G} \in \mathbb{R}^{n_g}$，其中 n_g 表示约束类型的数量。然而，这种表述不能被直接用于受约束优化，因为 $\mathcal{G}$ 包含无限多个不等式约束。因此，我们通过对违反约束的积分将 $\mathcal{G}$ 转化为有限维的形式。实际上，该积分被转化为采样惩罚函数的加权和 $J_\Sigma(\mathbf{c},\mathbf{T})$。在大多数约束解耦的情况下，即约束 $\mathcal{G}(p^{[s]}(t))$，其中 $t_i \le t < t_{i+1}$ 仅由 $\mathbf{c}_i$ 和 T_i, $p^{[s]} = [p, p^{(1)}, \cdots, p^{(s)}]$ 决定，某一轨迹片段的惩罚可以计算为

$$J_i(\mathbf{c}_i,T_i) = \frac{T_i}{\kappa_i}\sum_{j=0}^{\kappa_i} \bar{\omega}_j \chi^{\mathrm{T}} \max\left(\mathcal{G}(\mathbf{c}_i,T_i,\frac{j}{\kappa_i}),\mathbf{0}\right)^3, \tag{S9}$$

其中 κ_i 是第 i 片段上的样本编号， $\chi \in \mathbb{R}^{n_g}_{\ge 0}$ 是具有适当大值（比平滑惩罚大几个数量级）的惩罚权重向量，

and $(\bar{\omega}_0, \bar{\omega}_1, \ldots, \bar{\omega}_{\kappa_i-1}, \bar{\omega}_{\kappa_i}) = (1/2, 1, \cdots, 1, 1/2)$ the quadrature coefficients following the trapezoidal rule. We define the points determined by $\{\mathbf{c}_i, T_i, j/\kappa_i\}$ as *constraint points* $\mathring{\mathbf{p}}_{i,j} := p_i((j/\kappa_i)T_i)$, with $p_i(t)$ the i-th polynomial piece. Then $\mathcal{G}(\mathring{\mathbf{p}}_{i,j}) := \mathcal{G}(\mathbf{c}_i, T_i, j/\kappa_i)$. Note that $\mathring{\mathbf{p}}_{i,\kappa_i} = \mathring{\mathbf{p}}_{i+1,0}$. The cubic penalty is used here for its twice continuous differentiability. $J_\Sigma(\mathbf{c}, \mathbf{T})$ is then defined as the summation:

$$J_\Sigma(\mathbf{c}, \mathbf{T}) = \sum_{i=1}^{M} J_i(\mathbf{c}_i, T_i). \tag{S10}$$

Since the gradients w.r.t. $(\mathbf{c}, \mathbf{T})$ are constructed from all $(\mathbf{c}_i, \mathbf{T}_i)$, we only give gradient templates for $\mathbf{c}_i$ and T_i using the chain rule when Eq. S9 holds,

$$\frac{\partial J_\Sigma}{\partial \mathbf{c}_i} = \frac{\partial J_i}{\partial \mathbf{c}_i} = \frac{\partial J_i}{\partial \mathcal{G}} \frac{\partial \mathcal{G}}{\partial \mathbf{c}_i}, \tag{S11}$$

$$\frac{\partial J_\Sigma}{\partial T_i} = \frac{\partial J_i}{\partial T_i} = \frac{J_i}{T_i} + \frac{\partial J_i}{\partial \mathcal{G}} \frac{\partial \mathcal{G}}{\partial t} \frac{\partial t}{\partial T_i}, \tag{S12}$$

$$\frac{\partial J_i}{\partial \mathcal{G}} = 3\frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \chi^{\mathrm{T}} \max(\mathcal{G}, \mathbf{0})^2, \tag{S13}$$

$$t = \frac{j}{\kappa_i} T_i, \quad \frac{\partial t}{\partial T_i} = \frac{j}{\kappa_i}. \tag{S14}$$

From the above equations, Eqs. S11 –S14, once the gradients of constraints $\mathcal{G}$ relative to polynomial coefficients $\mathbf{c}_i$ and time t are evaluated, the gradients of $J_\Sigma(\mathbf{c}, \mathbf{T})$ are efficiently computed, which are then substituted into Eq. S8 to acquire final gradients respective to $\mathbf{q}$ and $\mathbf{T}$.

Penalty Formulation

The basic requirements on a desired trajectory are smoothness, total time, dynamical feasibility, as well as safety among obstacles and other agents. In this paper, we adopt MINCO to address above all concerns. Since MINCO is naturally guaranteed $\mathcal{C}^{s-1}$ by Theorem 2 in (*57*), no extra effort must be paid to trajectory continuity. The trajectory generation is formulated as an

以及 $(\bar{\omega}_0, \bar{\omega}_1, \ldots, \bar{\omega}_{\kappa_i-1}, \bar{\omega}_{\kappa_i}) = (1/2, 1, \cdots, 1, 1/2)$ 是遵循梯形规则的数值积分系数。我们定义由 $\{\mathrm{c}_i, T_i, j/\kappa_i\}$ 确定的点作为约束点 $\mathring{\mathrm{p}}_{i,j} := p_i((j/\kappa_i)T_i)$，其中 $p_i(t)$ 是第 i 个多项式片段。然后 $\mathcal{G}(\mathring{\mathrm{p}}_{i,j}) := \mathcal{G}(\mathrm{c}_i, T_i, j/\kappa_i)$。注意，$\mathring{\mathrm{p}}_{i,\kappa_i} = \mathring{\mathrm{p}}_{i+1}$，且为零。这里采用三次惩罚项是因为其具有二阶连续可微性。$J_\Sigma(\mathrm{c}, \mathrm{T})$ 被定义为以下求和：

$$J_\Sigma(\mathbf{c}, \mathbf{T}) = \sum_{i=1}^{M} J_i(\mathbf{c}_i, T_i). \tag{S10}$$

由于关于 $(\mathbf{c},\mathbf{T})$ 的梯度是由所有 $(\mathbf{c}_i, \mathbf{T}_i)$ 构建而成，我们仅在满足 Eq. S9 时，利用链式法则为$\mathbf{c}_i$ 和 T_i 提供梯度模板，

$$\frac{\partial J_\Sigma}{\partial \mathbf{c}_i} = \frac{\partial J_i}{\partial \mathbf{c}_i} = \frac{\partial J_i}{\partial \mathcal{G}} \frac{\partial \mathcal{G}}{\partial \mathbf{c}_i}, \tag{S11}$$

$$\frac{\partial J_\Sigma}{\partial T_i} = \frac{\partial J_i}{\partial T_i} = \frac{J_i}{T_i} + \frac{\partial J_i}{\partial \mathcal{G}} \frac{\partial \mathcal{G}}{\partial t} \frac{\partial t}{\partial T_i}, \tag{S12}$$

$$\frac{\partial J_i}{\partial \mathcal{G}} = 3\frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \chi^{\mathrm{T}} \max(\mathcal{G}, \mathbf{0})^2, \tag{S13}$$

$$t = \frac{j}{\kappa_i} T_i, \quad \frac{\partial t}{\partial T_i} = \frac{j}{\kappa_i}. \tag{S14}$$

由上述方程式，Eqs. S11 –S14，一旦对约束 $\mathcal{G}$ 关于多项式系数 $\mathbf{c}_i$ 和时间 t 的梯度被计算出来，$J_\Sigma(\mathbf{c},\mathbf{T})$ 的梯度就可以高效地计算出来，然后将其代入 Eq. S8，以获得相对于 $\mathbf{q}$ 和 $\mathbf{T}$ 的最终梯度。

惩罚项公式

对期望轨迹的基本要求包括平滑性、总时间、动力学可行性，以及在障碍物和其他代理之间的安全性。在本文中，我们采用MINCO方法来解决上述所有问题。由于MINCO在(*57*)中的定理2自然保证了 $\mathcal{C}^{s-1}$，因此无需额外努力来确保轨迹的连续性。轨迹生成被表述为一个

unconstrained optimization problem:

$$\min_{\mathbf{q},\mathbf{T}} \sum_{x} \lambda_x J_x, \tag{S15}$$

where J_x are J_Σ instances, λ_x is the weight for each penalty, and subscripts $x = \{s, t, d, o, w, u\}$ represent smoothness (s), total time (t), dynamical feasibility (d), obstacle avoidance (o), swarm reciprocal avoidance (w), and uniform distribution of constraint points (u), respectively. More task-specific penalties can also be added. The problem is solved via standard, unconstrained nonlinear optimization.

Smoothness J_s

The smoothness metric follows the definition in Eq. S3b. Without loss of generality, considering a single dimension of the i-th piece $p_i(t) : [0, T_i] \mapsto \mathbb{R}$ of a polynomial spline, its derivatives with respect to $\mathbf{c}_i$ and T_i are

$$\frac{\partial J_s}{\partial \mathbf{c}_i} = 2 \left(\int_0^{T_i} \beta^{(s)}(t) \beta^{(s)}(t)^{\mathrm{T}} dt \right) \mathbf{c}_i, \tag{S16}$$

$$\frac{\partial J_s}{\partial T_i} = \mathbf{c}_i^{\mathrm{T}} \beta^{(s)}(T_i) \beta^{(s)}(T_i)^{\mathrm{T}} \mathbf{c}_i. \tag{S17}$$

Total Time J_t

A shorter flight time is desirable, so we also minimize the weighted total execution time $J_t = \sum_{i=1}^{M} T_i$. Obviously, its gradients $\partial J_t / \partial \mathbf{c} = \mathbf{0}$, $\partial J_t / \partial \mathbf{T} = \mathbf{1}$.

Dynamical Feasibility Penalty J_d

For multicopters, dynamical feasibility is always guaranteed by limiting the trajectory's derivatives. In this paper, we limit the amplitude of velocity, acceleration, and jerk. Following Eq. S9, constraints of velocity, acceleration, and jerk are denoted by

$$\mathcal{G}_v = \dot{p}(t)^2 - v_m^2, \quad \mathcal{G}_a = \ddot{p}(t)^2 - a_m^2, \quad \mathcal{G}_j = \dddot{p}(t)^2 - j_m^2, \tag{S18}$$

无约束优化问题：

$$\min_{\mathbf{q},\mathbf{T}} \sum_{x} \lambda_x J_x, \tag{S15}$$

其中 J_x 是 J_Σ 实例， λ_x 是每个惩罚项的权重，下标 $x = \{s, t, d, o, w, u\}$ 分别代表平滑度（s）、总时间（t）、动力学可行性（d）、避障（o）、群体互避（w）以及约束点的均匀分布（u）。还可以添加更多特定任务的惩罚项。该问题通过标准的无约束非线性优化方法求解。

平滑度 J_s

平滑度指标遵循式 S3b 中的定义。在不失一般性的前提下，考虑多项式样条的第 i 段 $p_i(t)$： $[0, T_i] \mapsto \mathbb{R}$ ，其关于$\mathbf{c}_i$ 和 T_i 的导数为

$$\frac{\partial J_s}{\partial \mathbf{c}_i} = 2 \left(\int_0^{T_i} \beta^{(s)}(t) \beta^{(s)}(t)^{\mathrm{T}} dt \right) \mathbf{c}_i, \tag{S16}$$

$$\frac{\partial J_s}{\partial T_i} = \mathbf{c}_i^{\mathrm{T}} \beta^{(s)}(T_i) \beta^{(s)}(T_i)^{\mathrm{T}} \mathbf{c}_i. \tag{S17}$$

总时间 J_t

较短的飞行时间是理想的，因此我们也最小化加权的总执行时间 $J_t = \sum_{i=1}^{M} T_i$。显然，其梯度为 $\partial J_t / \partial \mathbf{c} = \mathbf{0}$， $\partial J_t / \partial \mathbf{T} = \mathbf{1}$。

动力学可行性惩罚 J_d

对于多旋翼飞行器，动态可行性总是通过限制轨迹的导数来保证。在本文中，我们限制速度、加速度和跃度的幅值。根据式子 S9，速度、加速度和跃度的约束表示为

$$\mathcal{G}_v = \dot{p}(t)^2 - v_m^2, \quad \mathcal{G}_a = \ddot{p}(t)^2 - a_m^2, \quad \mathcal{G}_j = \dddot{p}(t)^2 - j_m^2, \tag{S18}$$

where v_m, a_m, j_m are maximum allowed velocity, acceleration and jerk, respectively. The corresponding gradients are

$$\frac{\partial \mathcal{G}_x}{\partial \mathbf{c}_i} = 2\beta^{(n)}(t)p_i^{(n)}(t)^{\mathrm{T}}, \quad \frac{\partial \mathcal{G}_x}{\partial t} = 2\beta^{(n+1)}(t)^{\mathrm{T}}\mathbf{c}_i p_i^{(n)}(t), \tag{S19}$$

where $x = \{v, a, j\}$, $n = \{1, 2, 3\}$, and $t = jT_i/\kappa_i$, respectively. Substituting Eq. S18 into S9 and S19 into S11 & S12 obtains the penalty and the gradients of $\mathbf{c}_i$ and $\mathbf{T}_i$.

Obstacle Avoidance Penalty J_o

In this paper, we adopt the front-end of collision evaluation devised by (*14*). A brief description is given here for better understanding. In (*14*), the information for an unsafe trajectory to escape collision is extracted by comparing the unsafe initial trajectory with a collision-free guiding path. As depicted in Fig. S6, a fixed safe point $\mathbf{s}$ with a fixed safe vector $\mathbf{v}$ is generated and recorded by a corresponding key point $\mathbf{p}_{\mathrm{key}}$ on the trajectory. Then, the distance of $\mathbf{p}_{\mathrm{key}}$ to the obstacle that the $\{\mathbf{s}, \mathbf{v}\}$ pair generated is defined as

$$d_o(\mathbf{p}_{\mathrm{key}}, \{\mathbf{s}, \mathbf{v}\}) = (\mathbf{p}_{\mathrm{key}} - \mathbf{s})^{\mathrm{T}}\mathbf{v}. \tag{S20}$$

Using the distance information, the trajectory deforms iteratively during optimization. If $\mathbf{p}_{\mathrm{key}}$ discovers a new obstacle, a new $\{\mathbf{s}, \mathbf{v}\}$ pair is stacked to $\mathbf{p}_{\mathrm{key}}$'s records. Therefore each $\mathbf{p}_{\mathrm{key}}$ may record several $\{\mathbf{s}, \mathbf{v}\}$ pairs. Note that each $\mathbf{p}_{\mathrm{key}}$ generates and possesses $\{\mathbf{s}, \mathbf{v}\}$ pairs privately, and thus does not calculate its distance from other $\mathbf{p}_{\mathrm{key}}$s' $\{\mathbf{s}, \mathbf{v}\}$ pairs.

In this paper, constraint points $\mathring{\mathbf{p}}_{i,j}$ with $1 \le i \le M, 1 \le j \le \kappa_i$ are selected as the key points $\mathbf{p}_{\mathrm{key}}$. To enforce safety, we penalize distance to obstacles by less than a obstacle clearance $\mathcal{C}_o$. Therefore, the obstacle avoidance constraint is defined as

$$\mathcal{G}_o = (\cdots, \mathcal{G}_{o_k}(\mathring{\mathbf{p}}_{i,j}), \cdots)^{\mathrm{T}} \in \mathbb{R}^{N_{\mathbf{sv}}}, \tag{S21}$$

$$\mathcal{G}_{o_k}(\mathring{\mathbf{p}}_{i,j}) = \mathcal{C}_o - d_o(\mathring{\mathbf{p}}_{i,j}, \{\mathbf{s}, \mathbf{v}\}_k), \tag{S22}$$

其中 v_m， a_m， j_m 分别是允许的最大速度、加速度和跃度。相应的梯度为

$$\frac{\partial \mathcal{G}_x}{\partial \mathbf{c}_i} = 2\beta^{(n)}(t)p_i^{(n)}(t)^{\mathrm{T}}, \quad \frac{\partial \mathcal{G}_x}{\partial t} = 2\beta^{(n+1)}(t)^{\mathrm{T}}\mathbf{c}_i p_i^{(n)}(t), \tag{S19}$$

其中 $x = \{v, a, j\}$、 $n = \{1, 2, 3\}$ 和 $t = jT_i/\kappa_i$，分别。将式 S18 代入式 S9，将式 S19 代入式 S11 和 S12，即得到$\mathbf{c}_i$ 的惩罚项及其梯度，以及 $\mathbf{T}_i$。

避障惩罚 J_o

本文采用由 (*14*) 提出的碰撞评估前端。为了更好理解，这里简要介绍。在 (*14*) 中，通过比较不安全的初始轨迹与无碰撞引导路径，提取出逃避碰撞的不安全轨迹信息。如图 S6 所示，生成并记录了一个固定的安全点 $\mathbf{s}$ 和一个固定的安全向量 $\mathbf{v}$ ，以及轨迹上的对应关键点 $\mathbf{p}^{\mathrm{key}}$ 。然后，定义了点 $\mathbf{p}^{\mathrm{key}}$ 到由 $\{\mathbf{s}$ 和 , $\mathbf{v}\}$ 生成的障碍物的距离：

$$d_o(\mathbf{p}_{\mathrm{key}}, \{\mathbf{s}, \mathbf{v}\}) = (\mathbf{p}_{\mathrm{key}} - \mathbf{s})^{\mathrm{T}}\mathbf{v}. \tag{S20}$$

利用距离信息，轨迹在优化过程中会迭代变形。如果 $\mathbf{p}^{\mathrm{key}}$ 发现新的障碍物，则会在 $\mathbf{p}^{\mathrm{key}}$ 的记录中堆叠一个新的 $\{\mathbf{s}$ 和 v 对。因此，每个 $\mathbf{p}^{\mathrm{key}}$ 可能会记录多个 $\{\mathbf{s}$ 和 v 对。注意，每个 $\mathbf{p}^{\mathrm{key}}$ 只会生成并拥有自己的 $\{\mathbf{s}$ 和 v 对，因此不会计算其与其他 $\mathbf{p}^{\mathrm{key}}$ 的 $\{\mathbf{s}$ 和 v 对的距离。

在本文中，约束点 $\mathring{\mathbf{p}}_{i,j}$ 与 $1 \le i \le M, 1 \le j \le \kappa_i$ 被选为关键点 $\mathbf{p}^{\mathrm{key}}$。为了确保安全，我们通过惩罚距离障碍物小于障碍物净空距离 $\mathcal{C}_o$ 来实现。因此，避障约束被定义为

$$\mathcal{G}_o = (\cdots, \mathcal{G}_{o_k}(\mathring{\mathbf{p}}_{i,j}), \cdots)^{\mathrm{T}} \in \mathbb{R}^{N_{\mathbf{sv}}}, \tag{S21}$$

$$\mathcal{G}_{o_k}(\mathring{\mathbf{p}}_{i,j}) = \mathcal{C}_o - d_o(\mathring{\mathbf{p}}_{i,j}, \{\mathbf{s}, \mathbf{v}\}_k), \tag{S22}$$

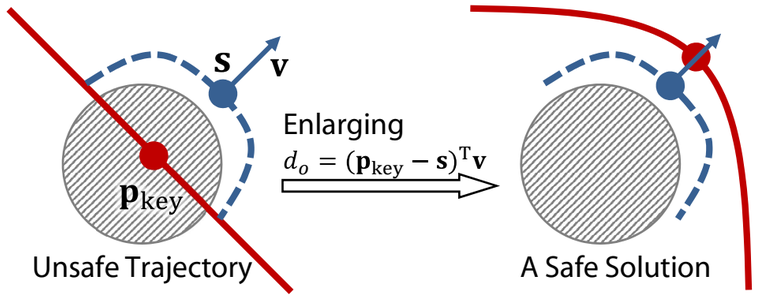


Fig. S6. Collision avoidance formulation. The red curve is the optimized trajectory and blue curve a collision-free path that starts and ends on the red one. A fixed point $\mathbf{s}$ with a vector $\mathbf{v}$ is generated to determine the collision penalty.

where $N_{\mathbf{sv}}$ is the number of $\{\mathbf{s}, \mathbf{v}\}$ pairs recorded by a single $\mathring{\mathbf{p}}_{i,j}$. For the case in which $d_o(\mathring{\mathbf{p}}_{i,j}, \{\mathbf{s}, \mathbf{v}\}_k) \leq \mathcal{C}_o$, the gradient is computed as

$$\frac{\partial \mathcal{G}_{o_k}}{\partial \mathbf{c}_i} = -\beta(t)\mathbf{v}^{\mathrm{T}}, \quad \frac{\partial \mathcal{G}_{o_k}}{\partial t} = -\dot{p}_i(t)^{\mathrm{T}}\mathbf{v}. \tag{S23}$$

Otherwise, $\partial \mathcal{G}_{o_k}/\partial \mathbf{c}_i = \mathbf{0}_{2s\times m}, \partial \mathcal{G}_{o_k}/\partial t = 0$.

Swarm Reciprocal Avoidance Penalty J_w

In this paper, a decentralized, non-cooperative swarm framework is adopted, which means that one agent receives other agents' trajectories as constraints and generates trajectories only for itself. Considering the u-th agent in a multicopter swarm containing U agents, swarm collision avoidance is guaranteed when the trajectory ${}^{u}p(t)$ of agent u maintains a distance greater than a safe clearance $\mathcal{C}_w$ to all the trajectories at the same global timestamps of other agents as depicted in Fig. S7. However, caution must be taken because we have always been using a relative time $t = jT_i/\kappa_i$ for $p_i(t)$ in our optimization, while the other agents' trajectories take an absolute timestamp $\tau = t_{of} + T_1 + \cdots + T_{i-1} + jT_i/\kappa_i$, with $t_{of} = t_w - \tau_w$ as a time offset to align t and τ from different world time t_w and τ_w, respectively. This indeed makes the penalty evaluated on the i-th piece depend on all its preceding piece times. We denote by

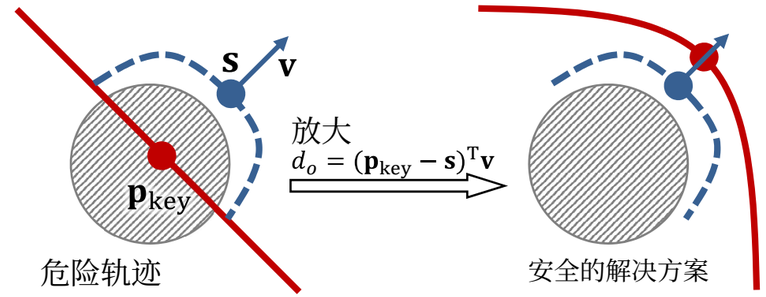


图 S6. 避碰公式。红色曲线为优化轨迹，蓝色曲线为起点和终点都在红色轨迹上的无碰撞路径。生成一个固定点$\mathbf{s}$，并带有向量$\mathbf{v}$，以确定碰撞惩罚。

其中 $N_{\mathbf{sv}}$ 是由单个 $\mathring{\mathbf{p}}_{i,j}$ 记录的 $\{\mathbf{s},\mathbf{v}\}$ 对的数量。对于 $d_o(\mathring{\mathbf{p}}_{i,j}, \{\mathbf{s},\mathbf{v}\}_k) \leq \mathcal{C}_o$ 的情况，梯度计算如下：

$$\frac{\partial \mathcal{G}_{o_k}}{\partial \mathbf{c}_i} = -\beta(t)\mathbf{v}^{\mathrm{T}}, \quad \frac{\partial \mathcal{G}_{o_k}}{\partial t} = -\dot{p}_i(t)^{\mathrm{T}}\mathbf{v}. \tag{S23}$$

否则， $\partial \mathcal{G}_{o_k}/\partial \mathbf{c}_i = \mathbf{0}_{2s\times m}, \partial \mathcal{G}_{o_k}/\partial t = 0$。

蜂群互避惩罚 J_w

本文采用去中心化的非合作群体框架，这意味着一个智能体将其他智能体的轨迹作为约束条件，并仅为自身生成轨迹。考虑包含 U 个智能体的多旋翼群体中的第 u个智能体，当智能体 u 的轨迹 ${}^{u}p(t)$在所有其他智能体在相同全局时间戳下的轨迹保持大于安全间隙 $\mathcal{C}_w$ 的距离时，群体碰撞避免得以保证，如图S7所示。然而，必须注意的是，我们在优化中一直使用相对时间 $t = jT_i/\kappa_i$ 来表示 $p_i(t)$，而其他智能体的轨迹采用绝对时间戳 $\tau = t_{of} + T_1 + \cdots + T_{i-1} + jT_i/\kappa_i$，其中 $t_{of} = t_w - \tau_w$作为时间偏移，用于对齐 t 和 τ，它们分别来自不同的世界时间 t_w 和 τ_w。这确实导致在第 i段上评估的惩罚依赖于其所有前置段的时间。我们用

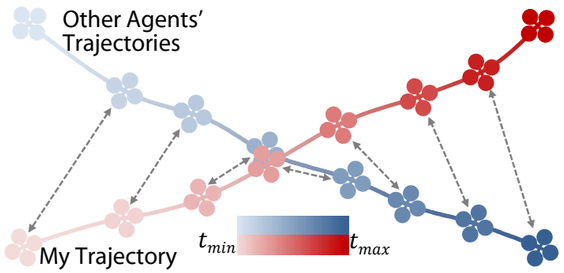


Fig. S7. Reciprocal collision avoidance formulation. Gray dotted arrows indicate states at the same absolute time, at which a safe distance must be ensured.

$G_w^{i,j}$ the constraint evaluated at the j-th constraint point on the i-th piece of ${}^u p(t)$, i.e., ${}^u p_i(t)$.

Therefore the reciprocal avoidance constraint is defined as

$$\mathcal{G}_w^{i,j}({}^u p_i(t), \tau) = (\cdots, \mathcal{G}_{w_k}({}^u p_i(t), \tau), \cdots)^{\mathrm{T}} \in \mathbb{R}^{U-1}, k \neq u, \tag{S24}$$

$$\mathcal{G}_{w_k}^{i,j}({}^u p_i(t), \tau) = \mathcal{C}_w^2 - d_w^2({}^u p_i(t), {}^k p(\tau)), \tag{S25}$$

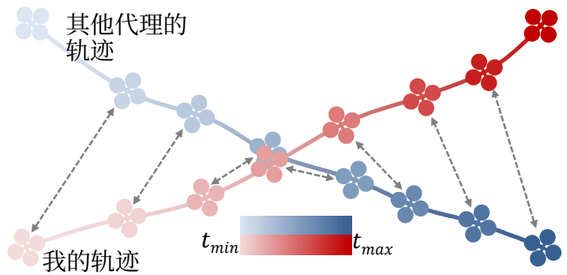
$$d_w({}^u p_i(t), {}^k p(\tau)) = \left\| \mathbf{E}^{1/2}({}^u p_i(t) - {}^k p(\tau)) \right\|, \tag{S26}$$

where ${}^k p(\tau)$ is the trajectory of the k-th agent that the u-th agent must avoid, $k \neq u$. The matrix $\mathbf{E} := \mathrm{diag}(1, 1, 1/c)$ with $c > 1$ transforms Euclidean distance into *ellipsoidal distance* with the minor axes at the z-axis to relieve the downwash risk from rotors. Squared distance is used to avoid *square root* operations.

When $d_w^2({}^u p_i(t), {}^k p(\tau)) \leq \mathcal{C}_w^2$, the gradient to $\mathbf{c}_i$ is

$$\frac{\partial \mathcal{G}_{w_k}^{i,j}}{\partial \mathbf{c}_i} = -2\beta(t)({}^u p_i(t) - {}^k p(\tau))^{\mathrm{T}} \mathbf{E}. \tag{S27}$$

The gradient to $\mathbf{T}$ is more complicated, since τ is a function of not only T_i, but T_1 to T_{i-1} as well when calculating the gradients for $\mathcal{G}_{w_k}^{i,j}$. This led to the gradient-propagate template Eq. S12 not holding here for $\partial J_\Sigma / \partial T_i \neq \partial J_i / \partial T_i$. When $d_w^2({}^u p_i(t), {}^k p(\tau)) \leq \mathcal{C}_w^2$, considering both preceding and current time profiles T_l for $1 \leq l \leq i$, the proper gradient should be computed as



图S7. 互反避碰公式。 灰色虚线箭头表示在相同绝对时间点的状态，必须确保安全距离。

$G^{i,}{}_w{}^j$在第 i段的第 j个约束点上评估的约束，即 ${}^u p_i(t)$。

因此，互反避让约束定义为

$$\mathcal{G}_w^{i,j}({}^u p_i(t), \tau) = (\cdots, \mathcal{G}_{w_k}({}^u p_i(t), \tau), \cdots)^{\mathrm{T}} \in \mathbb{R}^{U-1}, k \neq u, \tag{S24}$$

$$\mathcal{G}_{w_k}^{i,j}({}^u p_i(t), \tau) = \mathcal{C}_w^2 - d_w^2({}^u p_i(t), {}^k p(\tau)), \tag{S25}$$

$$d_w({}^u p_i(t), {}^k p(\tau)) = \left\| \mathbf{E}^{1/2}({}^u p_i(t) - {}^k p(\tau)) \right\|, \tag{S26}$$

其中 ${}^k p(\tau)$ 表示第 k 个智能体必须避免的第 u 个智能体的轨迹， $k \neq u$。矩阵$\mathbf{E}:=$ diag(1,1, $1/c$)，通过 $c > 1$ 转换欧几里得距离为椭圆距离，其短轴沿 z 轴，以减轻来自旋翼的下洗风险。采用平方距离以避免 开方运算。

当 $d_w^2({}^u p_i(t)\ {}^k p(\tau)) \leq \mathcal{C}_w^2$ 时，关于 $\mathbf{c}_i$ 的梯度为

$$\frac{\partial \mathcal{G}_{w_k}^{i,j}}{\partial \mathbf{c}_i} = -2\beta(t)({}^u p_i(t) - {}^k p(\tau))^{\mathrm{T}} \mathbf{E}. \tag{S27}$$

梯度对于 $\mathbf{T}$ 的导数更为复杂，因为 τ 不仅是 T_i 的函数，在计算 $\mathcal{G}^{i,}{}_{w_k}{}^j$ 的梯度时，T_1 到 T_{i-1} 也会影响。这导致了梯度传播模板方程式 Eq. S12 在这里对于 $\partial J_\Sigma / \partial T_i \neq \partial J_i / \partial T_i$ 不成立。当 $d_w^2({}^u p_i(t)\ {}^k p(\tau)) \leq \mathcal{C}_w^2$ 时，考虑到前一时刻和当前时刻的两个时间轮廓 T_l，用于 $1 \leq l \leq i$，正确的梯度应被计算为:

$$\frac{\partial J_w}{\partial T_l} = \sum_{k=1,k\neq u}^{U} \frac{\partial J_{w_k}}{\partial T_l} = \sum_{k=1,k\neq u}^{U} \sum_{i=l}^{M} \sum_{j=1}^{\kappa_i} \frac{\partial J^{i,j}_{w_k}}{\partial T_l}, \tag{S28}$$

$$\frac{\partial J^{i,j}_{w_k}}{\partial T_l} = \frac{J^{i,j}_{w_k}}{T_l} + \frac{\partial J^{i,j}_{w_k}}{\partial \mathcal{G}^{i,j}_{w_k}} \frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial T_l}, \tag{S29}$$

$$\frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial T_l} = \frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial t} \frac{\partial t}{\partial T_l} + \frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial \tau} \frac{\partial \tau}{\partial T_l}, \tag{S30}$$

$$\frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial t} = 2\left({}^{k}p(\tau) - {}^{u}p_i(t)\right)^{\mathrm{T}} \mathbf{E}^{u}\dot{p}_i(t), \tag{S31}$$

$$\frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial \tau} = 2\left({}^{u}p_i(t) - {}^{k}p(\tau)\right)^{\mathrm{T}} \mathbf{E}^{k}\dot{p}(\tau), \tag{S32}$$

$$\frac{\partial t}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i} & l = i, \\ 0 & l < i, \end{cases} \qquad \frac{\partial \tau}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i} & l = i, \\ 1 & l < i. \end{cases} \tag{S33}$$

The trajectory ${}^{k}p(\tau)$ for any other agent is already known, and thus its derivatives are all available. $\partial J^{i,j}_{w_k}/\partial \mathcal{G}^{i,j}_{w_k}$ in Eq. S29 is computed following Eq. S13, while $(\partial \mathcal{G}/\partial t)(\partial t/\partial T_i)$ in Eq. S12 is replaced by Eq. S30. Next $\partial J_w/\partial \mathbf{T} = [\partial J_w/\partial T_1, \cdots, \partial J_w/\partial T_M]^{\mathrm{T}}$.

Uniform Distribution Penalty J_u

This penalty is not mentioned in the main body of the article since this well-packed module will not bring about sufficient insights to general readers in terms of understanding the methodology, but only some confusion. The purpose of this penalty is to make the constraint points $\mathring{\mathbf{p}}_{i,j}$ equally spaced for all $1 \leq i \leq M$ and $0 < j \leq \kappa_i$, which is a simpler substitution to the space-uniform variant of MINCO mentioned in (*57*). It is necessary since if all MINCO parameters $\mathbf{q}$ and $\mathbf{T}$ are optimized freely, some pieces of the trajectory tend to vanish to reduce the total cost. This phenomenon is harmful for three reasons. First, $T_i = 0$ for the i-th trajectory piece is a unique singular point for MINCO, which does not consider consecutive aligned waypoints. Second, since the spatial collision avoidance is constrained by a finite number of constraint points, non-uniform distribution increases the possibility of skipping some tiny or thin obstacles, as shown in Fig. S8. Third, since we compute the distance to obstacles following Eq. S20, the

$$\frac{\partial J_w}{\partial T_l} = \sum_{k=1,k\neq u}^{U} \frac{\partial J_{w_k}}{\partial T_l} = \sum_{k=1,k\neq u}^{U} \sum_{i=l}^{M} \sum_{j=1}^{\kappa_i} \frac{\partial J^{i,j}_{w_k}}{\partial T_l}, \tag{S28}$$

$$\frac{\partial J^{i,j}_{w_k}}{\partial T_l} = \frac{J^{i,j}_{w_k}}{T_l} + \frac{\partial J^{i,j}_{w_k}}{\partial \mathcal{G}^{i,j}_{w_k}} \frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial T_l}, \tag{S29}$$

$$\frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial T_l} = \frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial t} \frac{\partial t}{\partial T_l} + \frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial \tau} \frac{\partial \tau}{\partial T_l}, \tag{S30}$$

$$\frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial t} = 2\left({}^{k}p(\tau) - {}^{u}p_i(t)\right)^{\mathrm{T}} \mathbf{E}^{u}\dot{p}_i(t), \tag{S31}$$

$$\frac{\partial \mathcal{G}^{i,j}_{w_k}}{\partial \tau} = 2\left({}^{u}p_i(t) - {}^{k}p(\tau)\right)^{\mathrm{T}} \mathbf{E}^{k}\dot{p}(\tau), \tag{S32}$$

$$\frac{\partial t}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i} & l = i, \\ 0 & l < i, \end{cases} \qquad \frac{\partial \tau}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i} & l = i, \\ 1 & l < i. \end{cases} \tag{S33}$$

任何其他代理的轨迹 ${}^{k}p(\tau)$ 已经已知，因此其导数全部可用。 $\partial J^{i,}_{w_k}{}^{j}/\partial \mathcal{G}^{i,}_{w_k}{}^{j}$在式 S29 中的计算遵循式 S13，而式 S12 中的 $(\partial \mathcal{G}/\partial t)(\partial t/\partial T_i)$ 被替换为式 S30。接下来 $\partial J_w/\partial \mathbf{T} = [\partial J_w/\partial T_1, \cdots, \partial J_w/\partial T_M]^{\mathrm{T}}$。

均匀分布惩罚 J_u

由于这个充分打包的模块不会在理解方法方面为一般读者带来足够的见解，而只会引起一些困惑，因此在正文中未提及此惩罚项。此惩罚的目的是使所有 $\mathring{\mathbf{p}}_{i,j}$的约束点在空间上 $1 \leq i \leq M$ 和 $0 < j \leq \kappa_i$上等距，这是一种比(*57*)中提到的空间均匀变体的MINCO更简单的替代方案。之所以有必要这样做，是因为如果所有的MINCO参数$\mathbf{q}$和$\mathbf{T}$ 被自由优化，一些轨迹段倾向于消失以降低总成本。这种现象有害，原因有三。第一， $T_i = 0$ 对于第 i段轨迹来说，是MINCO的唯一奇异点，不考虑连续对齐的路径点。第二，由于空间避障由有限数量的约束点限制，非均匀分布增加了跳过一些微小或狭窄障碍物的可能性，如图S8所示。第三，由于我们按照式（S20）计算到障碍物的距离，障碍物被建模为平面。当约束点沿平面移动且均匀分布减少此类移动时，这种建模的准确性会降低。

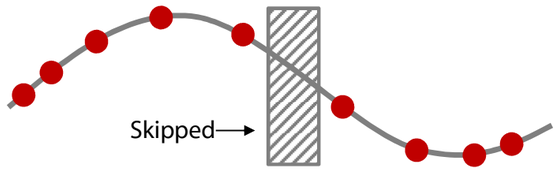


Fig. S8. Non-uniform constraint points (red dots) fail to detect a thin obstacle.

obstacles are modeled as planes. The accuracy of this modeling decreases when constraint points move along the plane and the uniform distribution reduces such movement.

To enforce the uniform distribution of constraint points, we penalize the variance of the squared distances between each pair of adjacent constraint points. For simplicity, we denote $N_c = \sum_{i=1}^{M} \kappa_i$ as the total number of distances. The variance is computed as

$$J_u = \mathrm{E}\left[\mathbf{R}^2\right] - \mathrm{E}\left[\mathbf{R}\right]^2 = \frac{1}{N_c}\left\|\mathbf{R}\right\|_2^2 - \frac{1}{N_c^2}\left\|\mathbf{R}\right\|_1^2, \tag{S34}$$

where $\mathbf{R} \in \mathbb{R}^{N_c}$ is defined by

$$\mathbf{R} = \left(\|\mathring{\mathbf{p}}_{1,1} - \mathring{\mathbf{p}}_{1,0}\|_2^2, \cdots, \|\mathring{\mathbf{p}}_{M,\kappa_M} - \mathring{\mathbf{p}}_{M,\kappa_M-1}\|_2^2\right). \tag{S35}$$

$\mathbf{R}$ is the squared distance vector for all $N_c + 1$ constraint points. We denote by $\mathbf{R}_k$ the k-th entry in $\mathbf{R}$, $k = j + \sum_{l=1}^{i-1} \kappa_l$. The gradient is computed as

$$\frac{\partial J_u}{\partial \mathbf{c}_i} = \sum_{j=1}^{\kappa_i} \beta\left(\frac{jT_i}{\kappa_i}\right)\left(\frac{\partial J_u}{\partial \mathring{\mathbf{p}}_{i,j}}\right)^{\mathrm{T}}, \tag{S36}$$

$$\begin{aligned}\frac{\partial J_u}{\partial \mathring{\mathbf{p}}_{i,j}} = &\frac{4}{N_c}\left(\mathbf{R}_{k-1} - \frac{\|\mathbf{R}\|_1}{N_c}\right)\left(\mathring{\mathbf{p}}_{i,j} - \mathring{\mathbf{p}}_{i,j-1}\right) \\ &- \frac{4}{N_c}\left(\mathbf{R}_k - \frac{\|\mathbf{R}\|_1}{N_c}\right)\left(\mathring{\mathbf{p}}_{i,j+1} - \mathring{\mathbf{p}}_{i,j}\right),\end{aligned} \tag{S37}$$

$$\frac{\partial J_u}{\partial T_i} = \frac{1}{\kappa_i}\sum_{j=1}^{\kappa_i} j\left(\frac{\partial J_u}{\partial \mathring{\mathbf{p}}_{i,j}}\right)^{\mathrm{T}} \mathbf{c}_i^{\mathrm{T}} \dot{\beta}\left(\frac{jT_i}{\kappa_i}\right). \tag{S38}$$

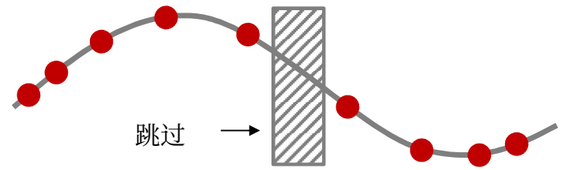


图 S8. 非均匀约束点（红点）无法检测到细长障碍物。

障碍物被建模为平面。当约束点沿平面移动且均匀分布减少此类移动时，这种建模的准确性会降低。

为了强制约束点的均匀分布，我们对每对相邻约束点之间的平方距离的方差进行惩罚。为简便起见，我们用$N_c = \sum_{i=1}^{M} \kappa_i$ 表示总距离数。方差的计算公式为：

$$J_u = \mathrm{E}\left[\mathbf{R}^2\right] - \mathrm{E}\left[\mathbf{R}\right]^2 = \frac{1}{N_c}\left\|\mathbf{R}\right\|_2^2 - \frac{1}{N_c^2}\left\|\mathbf{R}\right\|_1^2, \tag{S34}$$

其中$\mathbf{R} \in \mathbb{R}^{N_c}$ 由以下定义：

$$\mathbf{R} = \left(\|\mathring{\mathbf{p}}_{1,1} - \mathring{\mathbf{p}}_{1,0}\|_2^2, \cdots, \|\mathring{\mathbf{p}}_{M,\kappa_M} - \mathring{\mathbf{p}}_{M,\kappa_M-1}\|_2^2\right). \tag{S35}$$

$\mathbf{R}$ 是所有 $N_c + 1$ 约束点的平方距离向量。我们用$\mathbf{R}_k$ 表示$\mathbf{R}$中的第 k个条目，即 $k = j + \sum_{l=1}^{i-1} \kappa_l$。梯度的计算如下

$$\frac{\partial J_u}{\partial \mathbf{c}_i} = \sum_{j=1}^{\kappa_i} \beta\left(\frac{jT_i}{\kappa_i}\right)\left(\frac{\partial J_u}{\partial \mathring{\mathbf{p}}_{i,j}}\right)^{\mathrm{T}}, \tag{S36}$$

$$\begin{aligned}\frac{\partial J_u}{\partial \mathring{\mathbf{p}}_{i,j}} = &\frac{4}{N_c}\left(\mathbf{R}_{k-1} - \frac{\|\mathbf{R}\|_1}{N_c}\right)\left(\mathring{\mathbf{p}}_{i,j} - \mathring{\mathbf{p}}_{i,j-1}\right) \\ &- \frac{4}{N_c}\left(\mathbf{R}_k - \frac{\|\mathbf{R}\|_1}{N_c}\right)\left(\mathring{\mathbf{p}}_{i,j+1} - \mathring{\mathbf{p}}_{i,j}\right),\end{aligned} \tag{S37}$$

$$\frac{\partial J_u}{\partial T_i} = \frac{1}{\kappa_i}\sum_{j=1}^{\kappa_i} j\left(\frac{\partial J_u}{\partial \mathring{\mathbf{p}}_{i,j}}\right)^{\mathrm{T}} \mathbf{c}_i^{\mathrm{T}} \dot{\beta}\left(\frac{jT_i}{\kappa_i}\right). \tag{S38}$$

Temporal Constraint Elimination

An open-domain constraint is $\mathbf{T} \succ \mathbf{0}$, which is directly eliminated by a diffeomorphism map as is done in (*57*):

$$T_i = e^{\tau_i}, \tag{S39}$$

where τ_i is the unconstrained virtual time. Gradients w.r.t. T_i are propagated to τ_i following $\partial J/\partial \tau_i = (\partial J/\partial T_i)e^{\tau_i}$. More generally, any twice continuously differentiable bijection $f : \mathbb{R} \mapsto \mathbb{R}_{>0}$ suffices for this map.

Section S7. Collision Avoidance for Heterogeneous Swarms

In the conducted experiments, desired swarm clearance $\mathcal{C}_w$ is fixed due to identical agent size. If implementing the proposed planning algorithm for a heterogeneous swarm, each agent should broadcast its desired clearance, which is typically its radius plus a tolerance distance. Then, for the u-th agent that is going to avoid agent k, the swarm clearance between them is defined as ${}^{uk}\mathcal{C}_w = {}^{u}\mathcal{C}_w + {}^{k}\mathcal{C}_w$, where ${}^{u}\mathcal{C}_w$ and ${}^{k}\mathcal{C}_w$ are the desired clearances by agents u and k, respectively.

Section S8. Drone Removal on Depth Images

In the proposed system, an occupancy grid is adopted for static map representation, which can be affected by moving agents in depth images. Therefore, pixels of other agents are set to unknown. This procedure is performed at Line 14 in Alg. 1. The visual-inertial odometry that we utilize for localization requires grayscale images of static environments, which indeed contain moving agents. Nonetheless, the outlier is eliminated by RANSAC. Thus, moving agents are unlikely to affect our localization module which uses a wide angle-of-view camera.

时间约束消除

一个开放域约束是 $\mathbf{T} \succ \mathbf{0}$，它通过一个微分同胚映射直接消除，就像在 (*57*) 中所做的那样：

$$T_i = e^{\tau_i}, \tag{S39}$$

其中 τ_i 是无约束的虚拟时间。关于 T_i 的梯度会沿着 $\partial J/\partial \tau_i = (\partial J/\partial T_i)\, e^{\tau_i}$ 传播到 τ_i。更一般地，任何二次连续可微的双射 $f : \mathbb{R} \mapsto \mathbb{R}_{>0}$ 都足以用于此映射。

第 S7 节。异构群体的碰撞避免

在所进行的实验中，由于代理的尺寸相同，期望的群体间间隙 $\mathcal{C}_w$ 被固定。如果为异构群体实现所提出的规划算法，每个代理应广播其期望的间隙，通常为其半径加上容差距离。然后，对于第 u 个将要避免代理 k 的代理，定义它们之间的群体间隙为 ${}^{uk}\mathcal{C}_w = {}^{u}\mathcal{C}_w + {}^{k}\mathcal{C}_w$，其中 ${}^{u}\mathcal{C}_w$ 和 ${}^{k}\mathcal{C}_w$ 分别是代理 u 和 k 的期望间隙。

第 S8 节。深度图上的无人机移除

在所提出的系统中，采用占用网格进行静态地图的表示，这可能会受到深度图像中运动体的影响。因此，其他运动体的像素被设置为未知。这一过程在算法1的第14行执行。我们用于定位的视觉惯性测程需要静态环境的灰度图像，而这些图像实际上包含运动体。然而，异常值通过RANSAC被消除。因此，运动体不太可能影响我们使用广角相机的定位模块。

Algorithm 1: DroneDetection($\mathbf{u}, \mathbf{k}$)

Notation: Detector $\mathbf{u}$, Target Drone $\mathbf{k}$,
Detector Pose $\mathbf{P_u}$, Target Pose $\mathbf{P_k}$,
Depth Image $\mathbf{I_u}$, Threshold $\mathcal{T}$.
Input: $\mathbf{I_u}, \mathbf{P_u}, \mathbf{P_k}$
Output: $\mathbf{CorrectP_k}, \mathbf{ModifiedI_u}$
begin
  $\mathbf{PriorP_k} \leftarrow PoseWorld2Cam(\mathbf{P_k}, \mathbf{P_u})$
  $\mathbf{PriorPixel_k} \leftarrow Pose2Pixel(\mathbf{PriorP_k})$
  if $InDepthImage(\mathbf{PriorPixel_k}, \mathbf{I_u})$ **then**
    $\mathbf{Box} \leftarrow BoundingBox(\mathbf{PriorPixel_k}, \mathbf{I_u}, \mathbf{PriorP_k})$
    for Pixel in Box do
      $\mathbf{P} \leftarrow Pixel2Pose(\mathbf{Pixel})$
      if $Distance(\mathbf{P}, \mathbf{P_k}) \leq \mathcal{T}$ **then**
        $\mathbf{DronePixels_k}.PushBack(\mathbf{Pixel})$
    $\mathbf{CorrectPixel_k} \leftarrow GetCenter(\mathbf{DronePixels_k})$
    $\mathbf{CorrectP_k} \leftarrow Pixel2Pose(\mathbf{CorrectPixel_k})$
    $\mathbf{ModifiedI_u} \leftarrow SetUnknown(\mathbf{I_u}, \mathbf{DronePixels_k})$
  return;

Drone Detection

Unlike (*65*), in which drone detection is done merely by neural networks, we propose a lightweight but robust approach by pixel gathering from depth images using agent position as an a priori. The algorithm is explained in Alg. 1. Function *PoseWorld2Cam()* transforms $\mathbf{P_k}$ into the three-dimensional (3D) camera frame of the detector $\mathbf{u}$. Function *Pose2Pixel()* transforms $\mathbf{PriorP_k}$ into the 2D camera frame represented by the conventional $u - v$ pixel axis of the detector $\mathbf{u}$. Function *InDepthImage()* returns whether the pixel value of $\mathbf{PriorPixel_k}$ is within the boundary of the image $\mathbf{I_u}$. Function *BoundingBox()* creates a bounding box centered at $\mathbf{PriorPixel_k}$ with size larger if the z-component of $\mathbf{PriorP_k}$ on $\mathbf{I_u}$ is smaller. Function *Pixel2Pose()* transforms $\mathbf{Pixel}$ with depth from $\mathbf{I_u}$ into the 3D world frame. Function *Distance()* calculates Euclidean distance between $\mathbf{P}$ and $\mathbf{P_k}$. Function *PushBack()* means data saving. Function *Get-*

算法 1： DroneDetection($\mathbf{u}$,$\mathbf{k}$)

符号说明： 检测器 $\mathbf{u}$，目标无人机$\mathbf{k}$,
检测器姿态 $\mathbf{P_u}$，目标姿态$\mathbf{P_k}$,
深度图像 $\mathbf{I_u}$，阈值 $\mathcal{T}$。
输入： $\mathbf{I_u}$，$\mathbf{P_u}$，$\mathbf{P_k}$
输出： $\mathbf{CorrectP_k}$，$\mathbf{ModifiedI_u}$
开始
  $\mathbf{PriorP_k} \leftarrow PoseWorld2Cam(\mathbf{P_k}, \mathbf{P_u})$
  $\mathbf{PriorPixel_k} \leftarrow Pose2Pixel(\mathbf{PriorP_k})$
  如果 $InDepthImage(\mathbf{PriorPixel_k}, \mathbf{I_u})$ **那么**
    $\mathbf{Box} \leftarrow BoundingBox(\mathbf{PriorPixel_k}, \mathbf{I_u}, \mathbf{PriorP_k})$
    对盒子中的像素执行
      $\mathbf{P} \leftarrow Pixel2Pose(\mathbf{像素})$
      如果 $Distance(\mathbf{P}, \mathbf{P_k}) \leq \mathcal{T}$**那么**
        $\mathbf{DronePixels_k}.PushBack(\mathbf{像素})$
    $\mathbf{CorrectPixel_k} \leftarrow GetCenter(\mathbf{DronePixels_k})$
    $\mathbf{CorrectP_k} \leftarrow Pixel2Pose(\mathbf{CorrectPixel_k})$
    $\mathbf{ModifiedI_u} \leftarrow SetUnknown(\mathbf{I_u}, \mathbf{DronePixels_k})$
  return;

无人机检测

与（*65*）中仅通过神经网络进行无人机检测不同，我们提出了一种轻量但鲁棒的方法，利用先验的运动体位置从深度图像中采集像素。该算法在算法1中进行了说明。函数*PoseWorld2Cam()* 将$\mathbf{P_k}$ 转换为检测器的三维（3D）相机坐标系。函数*Pose2Pixel()* 将$\mathbf{PriorP_k}$转换为由检测器的传统 $u - v$ 像素轴表示的二维（2D）相机坐标系。函数*InDepthImage()* 返回$\mathbf{PriorPixel_k}$ 的像素值是否在图像$\mathbf{I_u}$的边界内。函数*BoundingBox()* 在$\mathbf{PriorPixel_k}$为中心创建一个边界框，如果$\mathbf{PriorP_k}$ 在$\mathbf{I_u}$ 中的z分量较小，则边界框的大小会更大。函数*Pixel2Pose()* 将带深度的像素从$\mathbf{Pixel}$ 转换到三维世界坐标系。函数*Distance()* 计算$\mathbf{P}$ 与$\mathbf{P_k}$之间的欧几里得距离。函数*PushBack()* 表示数据保存。函数*Get-*

Center() computes the average position of all the $\mathbf{DronePixels_k}$. Function *SetUnknown()* is an image operator that modifies the depth pixel values belonging to the detected drone to the unknown, and then the witnessed drone is removed. The $\mathbf{CorrectP_k}$ can be further used to correct localization drift.

Section S9. Failure Cases and Troubleshooting

Although a great deal of engineering considerations have been taken into account, limited by our knowledge and equipment, failures still occasionally occur. To those who are intended to reproduce our work to carry out their own research, possible solutions to some cases of failure are listed in Table S3.

Center() 计算所有$\mathbf{DronePixels_k}$的平均位置。函数*SetUnknown()* 是一种图像操作，用于将属于检测到的无人机的深度像素值修改为未知，然后移除被观察到的无人机。$\mathbf{CorrectP_k}$ 可以进一步用于校正定位漂移。

第S9节 失败案例与故障排除

尽管考虑了大量的工程因素，但由于我们的知识和设备的限制，故障仍然偶尔发生。对于那些打算复制我们的工作以进行自己研究的人，表S3中列出了一些可能的故障解决方案。

Table S3. Cases of failure and troubleshooting.

Failure Cases	Frequency	Possible Solutions
Arm denied by FCU.	Often if using ekf2 estimator in PX4.	Change the default ekf2 to Q estimator (complementary filter) in PX4 Autopilot.
Initial state estimation not stable.	Becomes common after a crash.	Re-calibrate extrinsic parameters between camera and IMU.
RealSense D430 failed to boot.	Often if the power supply is insufficient.	Make sure the type-c port has enough power and use a high-current type-c cable.
RealSense D430 outputs significantly wrong distance measurements.	Rare.	This happens when pointing the camera at a highlighted plane or a surface with repeated textures; therefore, try to add some features on these surfaces.
Some of the images are corrupted when in flight.	Rare.	Caused by signal interfering or vibration, so check the connection of the camera cable and place it away from the battery, ESC, and motors.
Trajectory message loss.	Sometimes.	Use better network hardware and do not print all the information to the ground computer via network.
Emergency stops triggered frequently in dense environments.	Sometimes.	Increase the map resolution and set appropriate inflation; otherwise, some small gaps frequently occur and continuously switch between free and occupied.
Two drones collide.	Rare.	Caused by the poor network every time and meanwhile, two drones did not see each other in depth images, so fix the network first and consider enlarging the depth FOV (field of view).
Hitting an obstacle.	Rare.	RealSense D430 is the culprit every time, so check what goes wrong with this device following the above troubleshooting regarding it.

Section S10. More Discussion on Communication

The requirements for communication are task-specific. In experiments *Fly Through Dense Forest*, the goal positions have been written into configuration files before takeoff. Then, trajec-

表S3. 失效案例与故障排查。

失效案例	频率	可能的解决方案
臂被FCU拒绝。	通常如果使用 ekf2 估算器 在 PX4 中	在 PX4 自动驾驶仪中，将默认的 ekf2 改为 Q 估算器（完全- 描述滤波器）。
初始状态估计 不稳定。	<style>/* 这里可以插入样式定义 */</style> 常见 崩溃后	重新校准之间的外在参数 相机和惯性测量单元 （IMU）。
RealSense D430 设备无法启动 。	通常如果 · 电源供应不足 。	确保Type-C端口有足够的电源，并 使用高电流Type-C线缆。
RealSense D430输出 明显错误 距离测量 。	· 罕见。	当将相机指向一个 高亮显示的平面或具有重复 纹理的表面时会发生这种情况；因此，尝试在 这些表面
部分图像是 损坏时在 飞行。	罕见。	由信号干扰或振动引起，因此 检查相机电缆的连接，并 将其远离电池、电子调速器（ESC）和电机。
轨迹信息丢失。	有时会发生。	使用更好的网络硬件，并且不要通过 将所有信息传输到地面计算机。 网络。
紧急停止 频繁触发 在密集环境中。	有时。	提高地图分辨率并设置合适的 通货膨胀；否则，一些小的间隙经常发生 发生并不断在空闲和 占用。
两架无人机碰撞。	罕见。	每次都由网络差引起，并且 同时 两架无人机未能互相看到 在深度图像中，因此首先固定网络，然后 考虑扩大深度视场（视野）。
遇到障碍。	罕见。	RealSense D430 每次都是罪魁祸首，因此 检查此设备出现问题的原因 按照上述关于它的故障排除进行检查。

第S10节。关于通信的更多讨论

通信的要求是任务特定的。在实验穿越密林中，目标位置已在起飞前写入配置文件中。然后，轨迹-

tory is the only message type to be transmitted for inter-robot collision avoidance and relative localization correction. In such cases, the communication range must be at least twice the planning distance, since drones with spacing closer than this threshold will affect each other when planning the trajectory. If their distance is longer than this threshold, they are sure to be collision-free in a short period of future time without any avoidance actions. Benefiting from the developments of wireless communication technology, such a dozens-of-meters communication range has been achieved by most solutions, e.g., WiFi, UWB, and Zigbee.

In formation flight tasks, except for reciprocal avoidance, drones have to dynamically maintain a formation according to other drones' real-time positions extracted from received trajectories. This requires that each drone maintains at least one connection to others and the whole formation is inseparable regarding to wireless connection. Then adopting some multihop and routing protocols, the trajectory messages can reach to all other drones. In the experiment *Formation Navigation in the Wild*, the formation scale is relatively compact, therefore messages are directly shared in the whole swarm.

In the experiment *Intensive Reciprocal Avoidance Evaluation*, trajectory-transmission requirements are the same as those in *Fly Through Dense Forest*. However, the goals in this experiment are given by a ground computer in real-time. Therefore, a connection from the ground computer to each drone is necessary. In addition, the ground computer can be connected directly as in our experiments or through wireless routing.

In the experiment *Multi-Drone Tracking with Target Occlusion*, each drone combines goals from its own observations with others' observations received from the network. Therefore, more sharing means higher robustness to occlusion. If there is no observation from others, the drone can still track the target, but may lose it when encountering a significant occlusion. Thus, it is difficult to determine an exact threshold of communication range in this task. In our experiment, we share the observation among all drones.

tory是用于机器人间避碰和相对定位校正唯一传输的消息类型。在这种情况下，通信范围必须至少是规划距离的两倍，因为当无人机之间的间距小于此阈值时，它们在规划轨迹时会相互影响。如果它们的距离超过此阈值，则在未来的短时间内，它们必然不会发生碰撞，无需采取避让措施。得益于无线通信技术的发展，这样数十米的通信范围已被大多数方案实现，例如WiFi、UWB和Zigbee。

在编队飞行任务中，除了相互避让外，无人机还必须根据接收轨迹中提取的其他无人机的实时位置，动态保持编队。这要求每架无人机至少与其他无人机保持一条连接，整个编队在无线连接方面是不可分割的。然后，采用一些多跳和路由协议，轨迹信息可以传达到所有其他无人机。在实验野外编队导航中，编队规模相对紧凑，因此信息可以直接在整个群体中共享。

在密集互避评估实验中，轨迹传输要求与穿越密林中的要求相同。然而，本次实验中的目标由地面计算机实时提供。因此，必须建立地面计算机与每个无人机之间的连接。此外，地面计算机可以像我们的实验一样直接连接，或通过无线路由进行连接。

在实验多无人机追踪与目标遮挡中，每架无人机将其自身观察到的目标信息与从网络接收的其他无人机的观察信息相结合。因此，更多的信息共享意味着对遮挡的鲁棒性更高。如果没有来自其他无人机的观察信息，无人机仍然可以追踪目标，但在遇到显著遮挡时可能会丢失目标。因此，在此任务中很难确定通信范围的确切阈值。在我们的实验中，我们在所有无人机之间共享观察信息。

Regarding communication bandwidth, in the bamboo forest experiment of 10 drones, the size of one trajectory is approximately 170 bytes. For the 65-m flight in the bamboo forest there are about 100 trajectories sent and 1000 trajectories received for each drone. The data rate (bandwidth) containing both sent and received information is approximately 2 kB/s and reached 8 kB/s at peak. These are minor values for WiFi network, which always provides a several megabytes-per-second (MBps) data rate. The statistics are listed in Table S4 and S5.

Table S4. Data sent in the bamboo forest flight. Trajectory is the only message type transmitted via network. **Traj.**: Trajectory; **Min.**: Minimal; **Max.**: Maximal; **Avg.**: Average.

Drone ID	Traj. Number	Total Size (bytes)	Min. Traj. Size (bytes)	Max. Traj. Size (bytes)	Avg. One Traj. Size (bytes)	Peak Data Rate (bytes/s)	Avg. Data Rate (bytes/s)
0	93	15811	127	175	170.01	875	171.63
1	96	16224	127	175	169.00	636	176.11
2	88	15064	127	175	171.18	525	163.52
3	98	16830	127	175	171.73	700	182.69
4	102	17418	127	175	170.76	1209	189.07
5	132	21628	127	175	163.85	2159	234.77
6	125	20755	127	175	166.04	1002	225.30
7	112	19088	127	175	170.43	2100	207.20
8	102	17514	127	175	171.71	700	190.12
9	93	15891	127	175	170.87	684	172.50
Avg.	**104.10**	**17622.30**	**127.00**	**175.00**	**169.56**	**1059.00**	**191.29**

关于通信带宽，在竹林实验中，10架无人机的轨迹数据大小大约为170字节。对于在竹林中进行的65米飞行，每架无人机需要发送大约100条轨迹，接收大约1000条轨迹。包含发送和接收信息的数据速率（带宽）大约为2 kB/s，峰值达到8 kB/s。这对于WiFi网络来说是微不足道的，后者通常提供每秒数兆字节（MBps）的数据传输速率。相关统计数据列在表S4和S5中。

表S4. 竹林飞行中的数据传输。 轨迹是通过网络传输的唯一消息类型。**轨迹**：Trajectory；**最小值**：Min.；**最大值**：Max.；**平均值**：Avg。

无人机编号	轨迹数量	总计大小（字节）	最小值 轨迹大小（字节）	最大值 轨迹大小（字节）	平均值 轨迹大小（字节）	峰值数据速率（字节/秒）	平均值 数据速率（字节/秒）
0	93	15811	127	175	170.01	875	171.63
1	96	16224	127	175	169.00	636	176.11
2	88	15064	127	175	171.18	525	163.52
3	98	16830	127	175	171.73	700	182.69
4	102	17418	127	175	170.76	1209	189.07
5	132	21628	127	175	163.85	2159	234.77
6	125	20755	127	175	166.04	1002	225.30
7	112	19088	127	175	170.43	2100	207.20
8	102	17514	127	175	171.71	700	190.12
9	93	15891	127	175	170.87	684	172.50
平均值	**104.10**	**17622.30**	**127.00**	**175.00**	**169.56**	**1059.00**	**191.29**

Table S5. Data received in the bamboo forest flight. Trajectory is the only message type transmitted via network. **Traj.**: Trajectory; **Min.**: Minimal; **Max.**: Maximal; **Avg.**: Average.

Drone ID	Traj. Number	Total Size (bytes)	Min. Traj. Size (bytes)	Max. Traj. Size (bytes)	Avg. One Traj. Size (bytes)	Peak Data Rate (bytes/s)	Avg. Data Rate (bytes/s)
0	948	160412	127	175	169.21	6202	1741.29
1	945	159999	127	175	169.31	5377	1736.80
2	953	161159	127	175	169.11	5838	1749.40
3	943	159393	127	175	169.03	5377	1730.22
4	939	158805	127	175	169.12	5043	1723.84
5	909	154595	127	175	170.07	4295	1678.14
6	916	155468	127	175	169.72	5377	1687.62
7	929	157135	127	175	169.14	4502	1705.71
8	939	158709	127	175	169.02	6538	1722.80
9	948	160332	127	175	169.13	5377	1740.42
Avg.	**936.90**	**158600.70**	**127.00**	**175.00**	**169.29**	**5392.60**	**1721.62**

表S5. 竹林飞行中接收的数据。 轨迹是通过网络传输的唯一消息类型。**轨迹**：Trajectory；**最小值**：Min.；**最大值**：Max.；**平均值**：Avg.。

无人机编号	轨迹数量	总计大小（字节）	最小值 轨迹大小（字节）	最大值 轨迹大小（字节）	平均值 轨迹大小（字节）	峰值数据速率（字节/秒）	平均值 数据速率（字节/秒）
0	948	160412	127	175	169.21	6202	1741.29
1	945	159999	127	175	169.31	5377	1736.80
2	953	161159	127	175	169.11	5838	1749.40
3	943	159393	127	175	169.03	5377	1730.22
4	939	158805	127	175	169.12	5043	1723.84
5	909	154595	127	175	170.07	4295	1678.14
6	916	155468	127	175	169.72	5377	1687.62
7	929	157135	127	175	169.14	4502	1705.71
8	939	158709	127	175	169.02	6538	1722.80
9	948	160332	127	175	169.13	5377	1740.42
平均值	**936.90**	**158600.70**	**127.00**	**175.00**	**169.29**	**5392.60**	**1721.62**

Figures

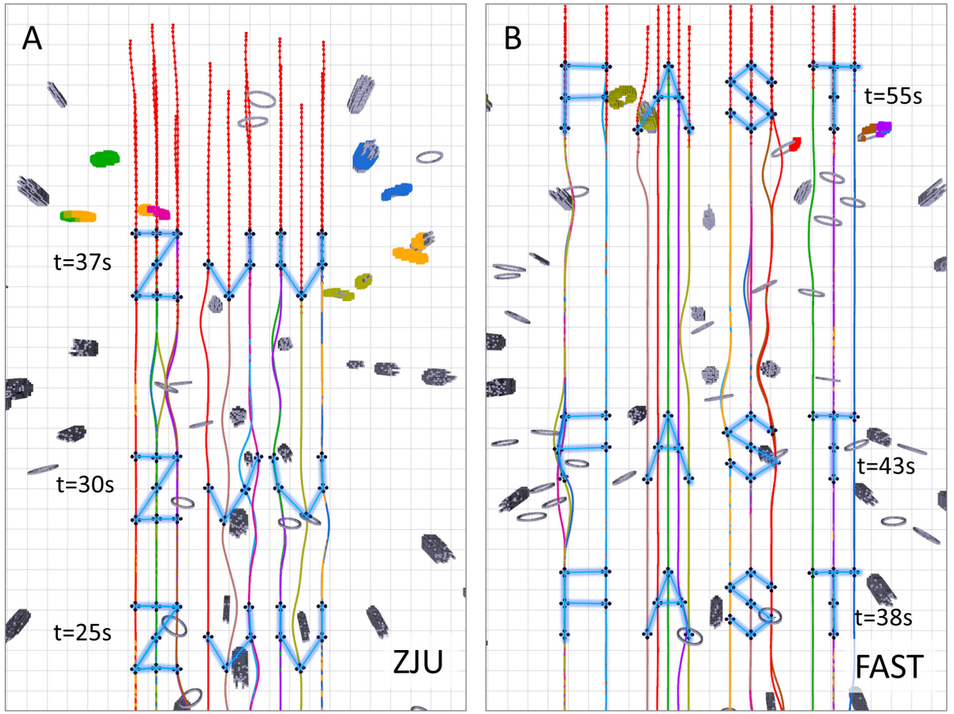
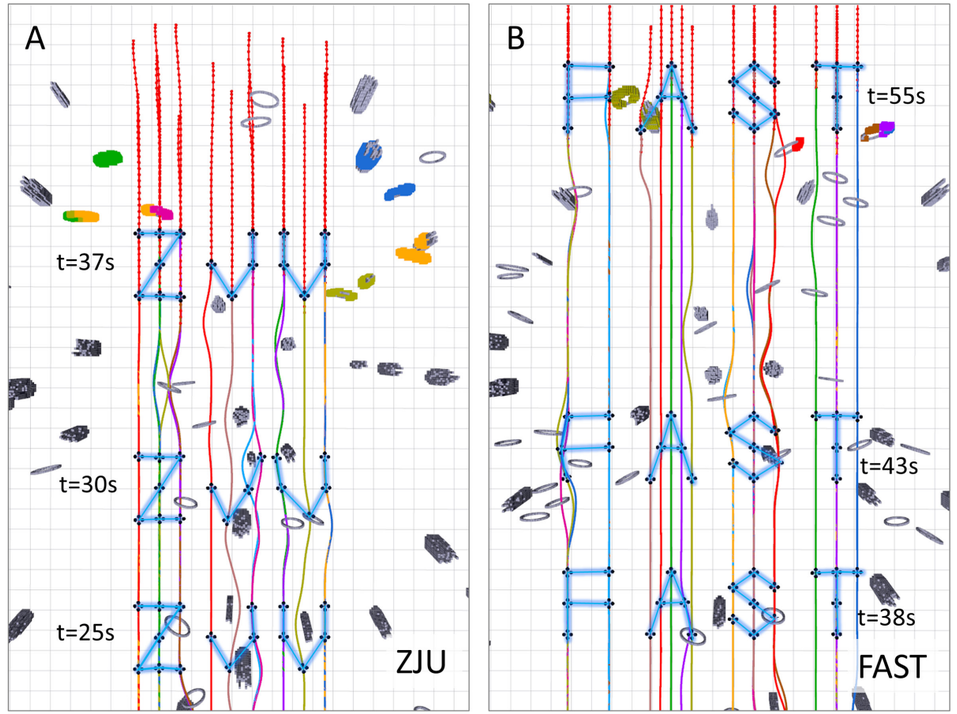


Fig. S9. Composite images of two formation flights. (**A**) Letters ZJU (ZheJiang University) formed by 16 drones. (**B**) Letters FAST (FAST Lab) formed by 22 drones.

图表



图S9. 两次编队飞行的复合图像。(**A**) 由16架无人机组成的“ZJU”（浙江大学）字样。(**B**) 由22架无人机组成的“FAST”（FAST实验室）字样。

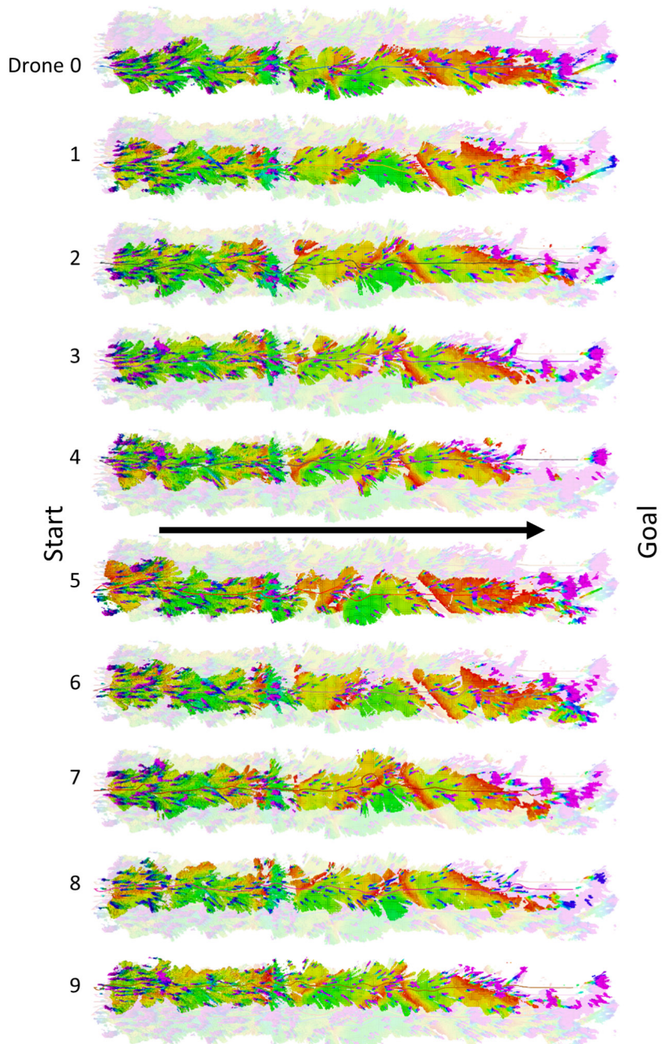


Fig. S10. Maps of the Bamboo Forest Built by Drones. The light-colored background denotes the map fused by overlaying 10 position-aligned drones' sub-maps. Different colors represent different heights. For example, green indicates the ground while the purple indicates bamboos. All trajectories are collision-free. Owing to the drift correction, 10 maps are all accurately aligned here.

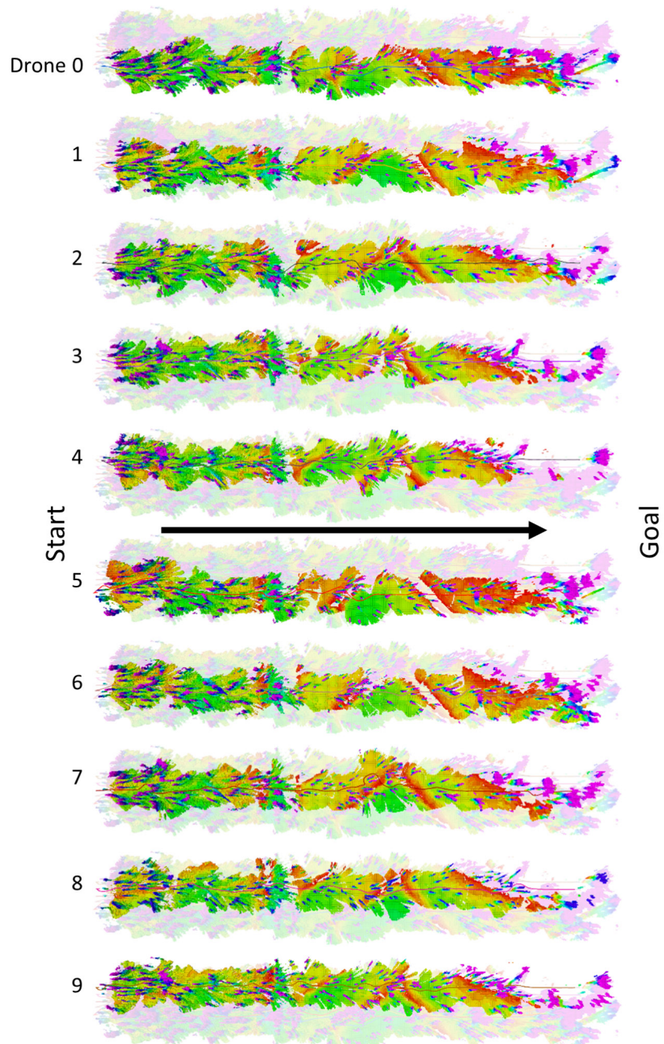


图 S10. 无人机构建的竹林地图 浅色背景表示通过叠加10个位置对齐的无人机子地图融合而成的地图。不同颜色代表不同的高度。例如，绿色表示地面，而紫色表示竹子。所有轨迹均无碰撞。由于漂移校正，所有10个地图在此都被精确对齐。

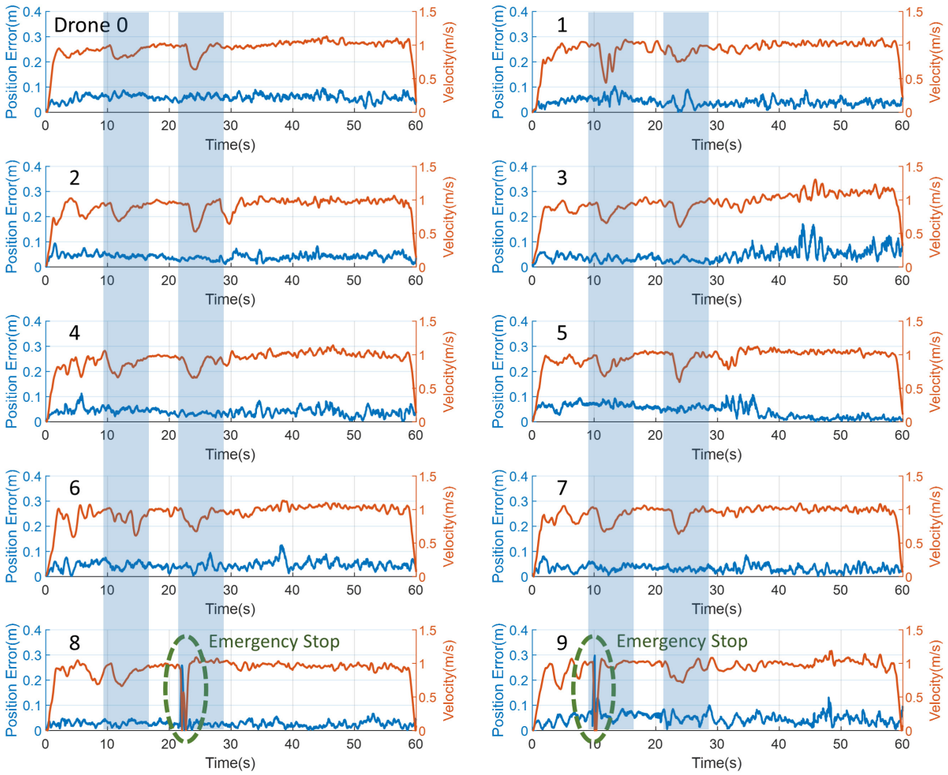


Fig. S11. Control errors during flights in Section *Formation Navigation in the Wild*. Green dotted ellipses indicate emergency stops activated by the safety check introduced in Section *Hierarchical Safety Guarantee*. Blue translucent areas indicate significant changes in speeds. Here, it is emergency stop rather than speed change that increases the position error.

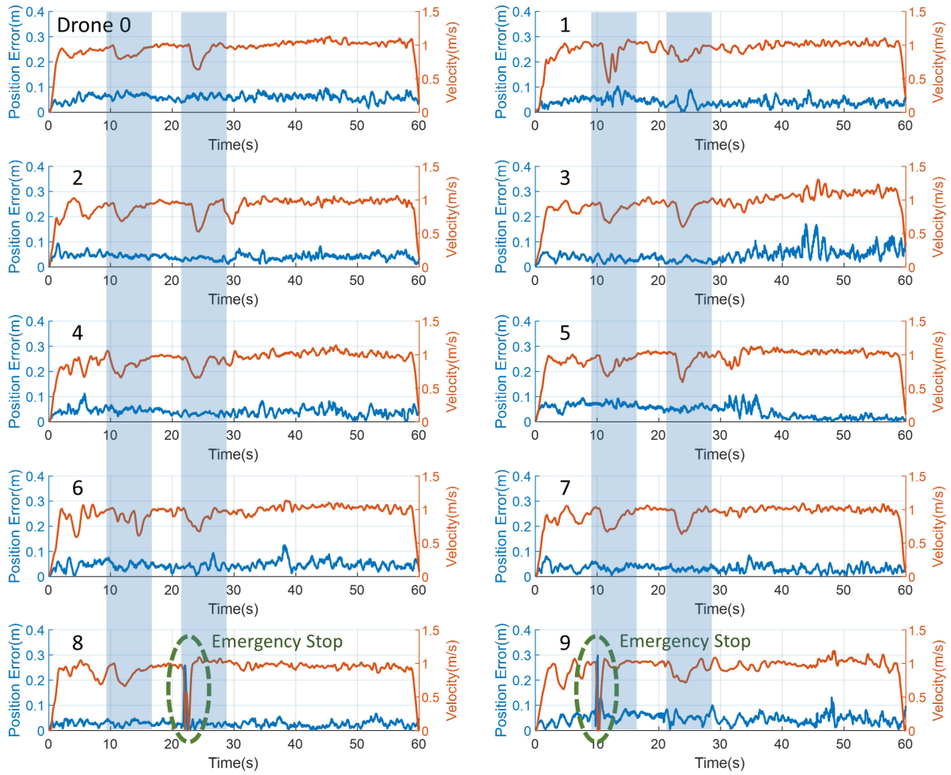


图 S11. 第节中的飞行控制误差野外编队导航**.** 绿色虚线椭圆表示由第层级安全保障节引入的安全检查触发的紧急停止。蓝色半透明区域表示速度的显著变化。这里，增加位置误差的是紧急停止而非速度变化。

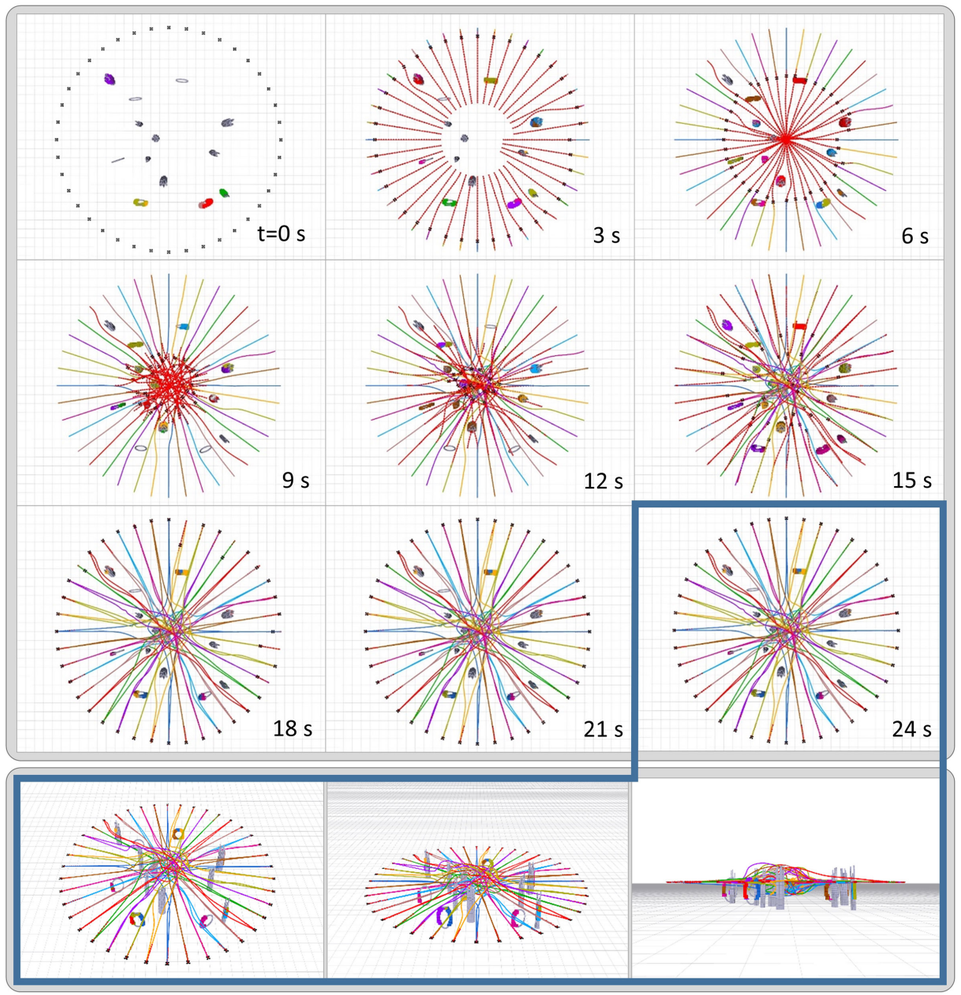


Fig. S12. A 40-drone position swap on a circle.

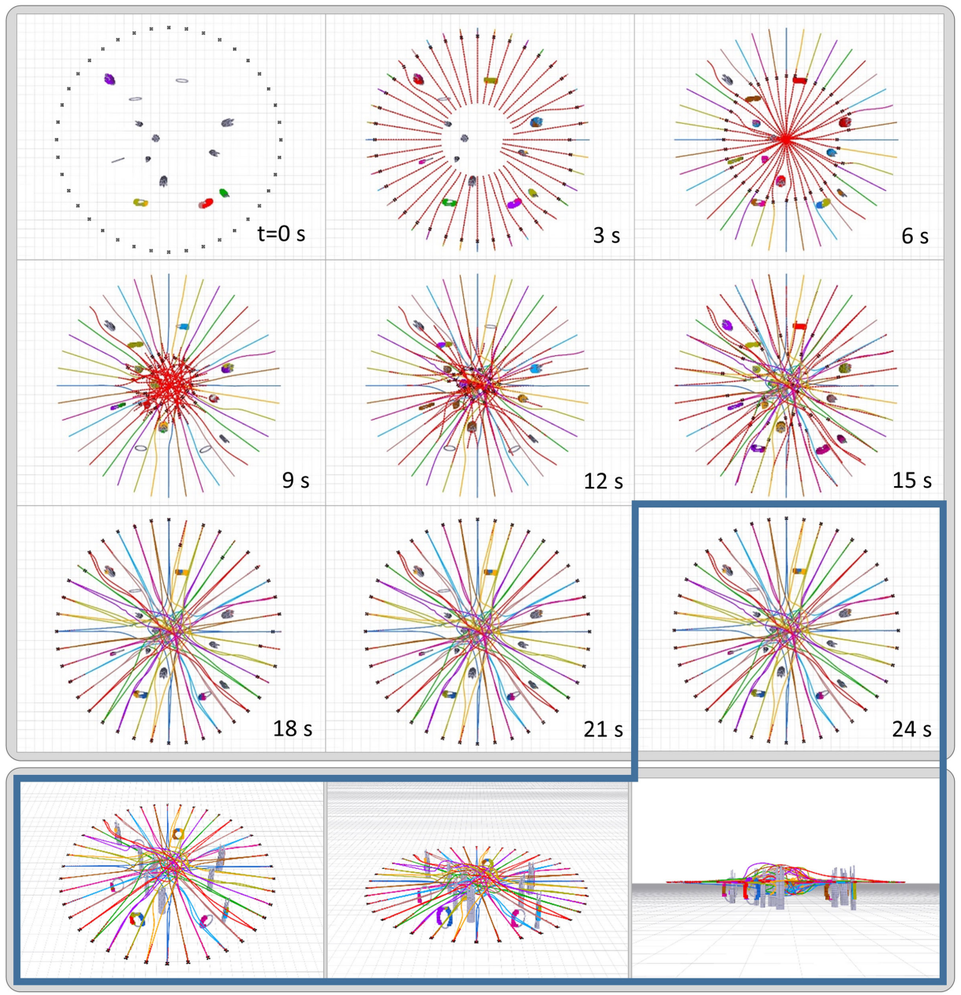


图 S12. 圆上的 40 架无人机位置交换。

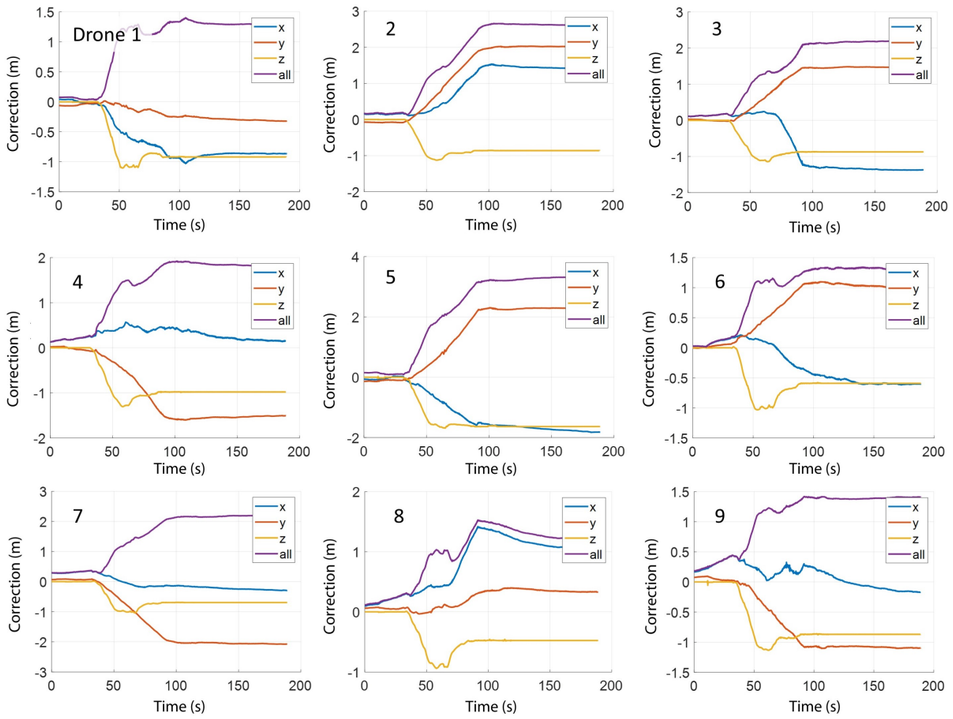


Fig. S13. Localization drift corrections for drones 1–9. Drone 0 is visualized in Fig. S3B.

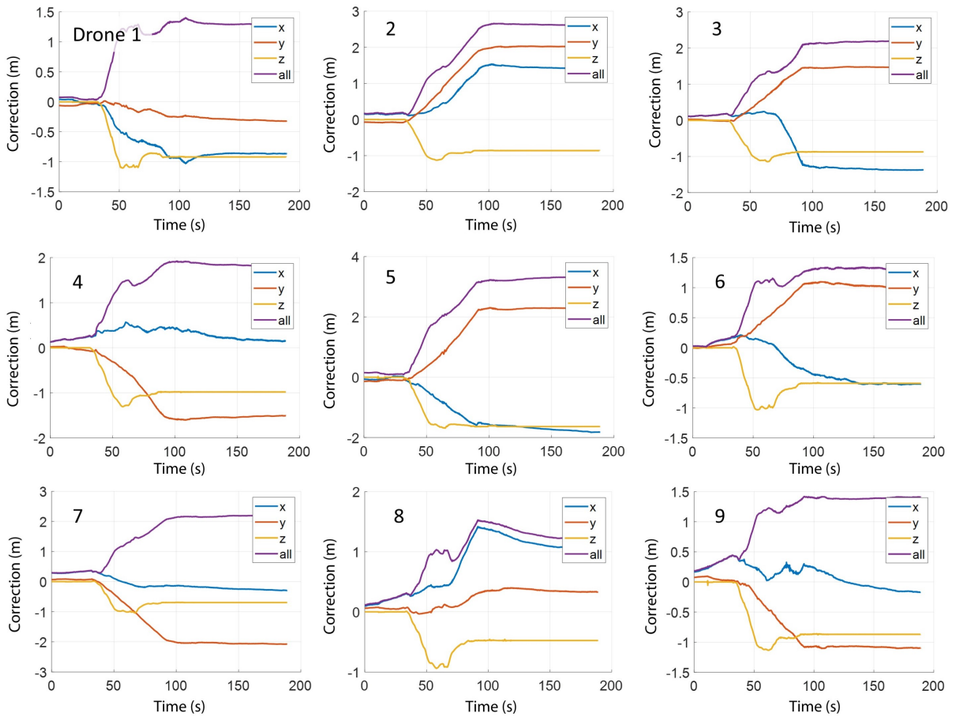


图 S13. 定位无人机 0 在图 S3B 中可视化

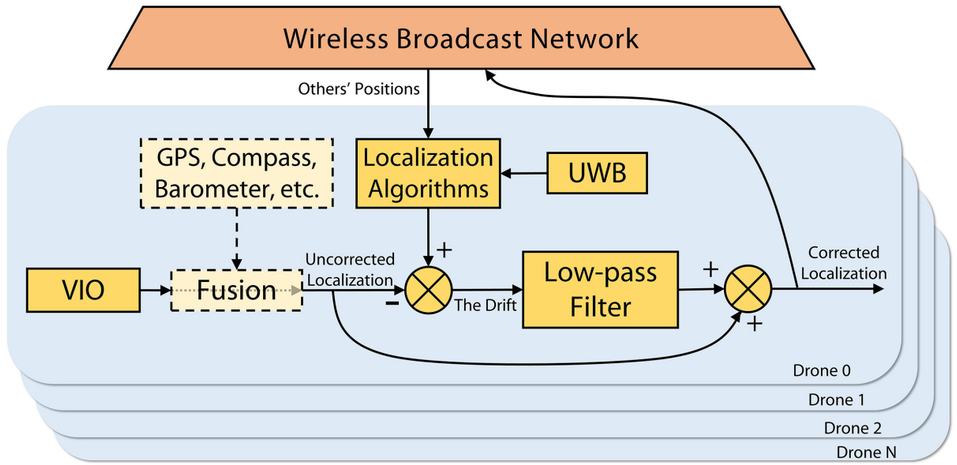


Fig. S14. Block diagram of localization drift correction. See Section *Localization and Drift Correction* for details. Sensors with corresponding fusion methods in dashed light-yellow boxes are optional and not adopted in our experiments. Positions in our framework are shared via trajectories.

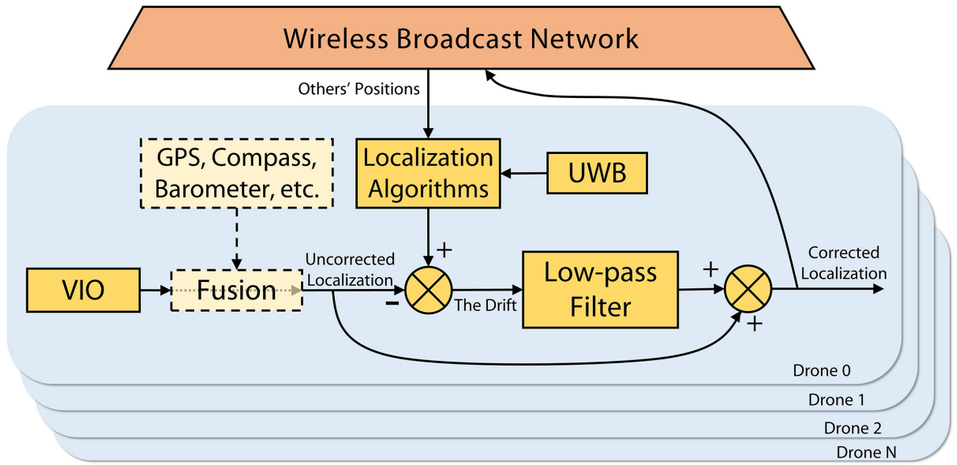


图 S14. 定位漂移校正的框图。 详见 定位与漂移校正 部分。虚线浅黄色框中的传感器及其对应的融合方法为可选项，在我们的实验中未采用。我们框架中的位置通过轨迹共享。

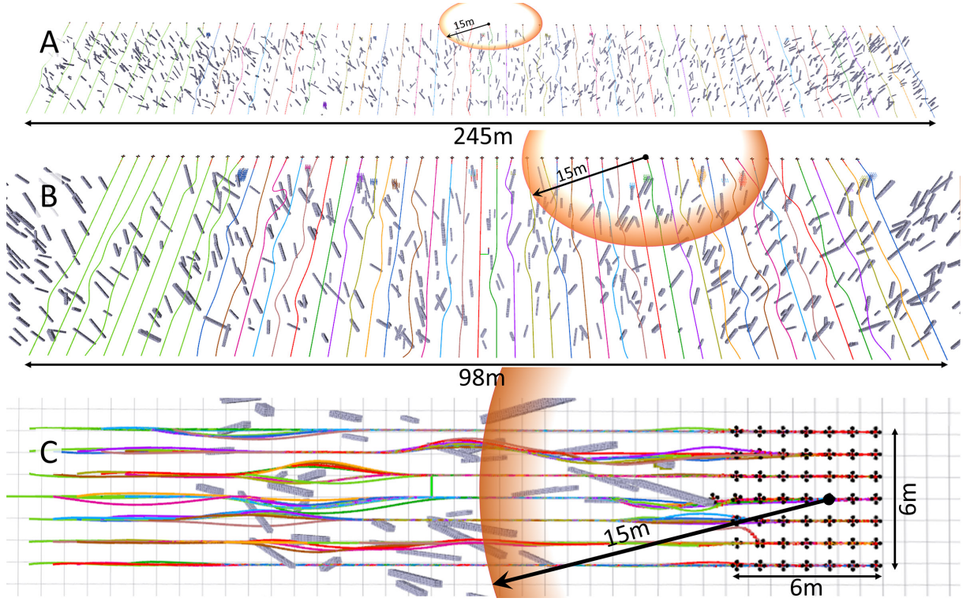
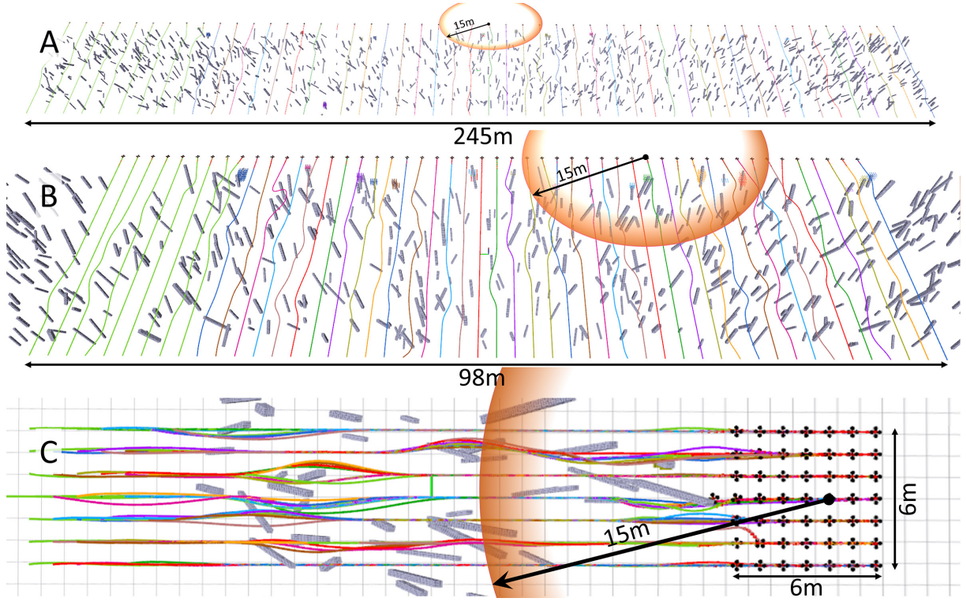


Fig. S15. Formation settings for scalability tests on the computation efficiency in Section *Time and Space Complexity*. (**A**) A loose formation. (**B**) A medium-intensity formation. (**C**) A tight formation. Orange circles indicate the range of two times the planning distance (7.5 m in Table S6).



图S15. 计算效率可扩展性测试的形成设置，见第时间与空间复杂性。 (**A**) 松散队形。 (**B**) 中等强度队形。 (**C**) 紧密队形。橙色圆圈表示两倍规划距离的范围（表S6中的7.5米）。

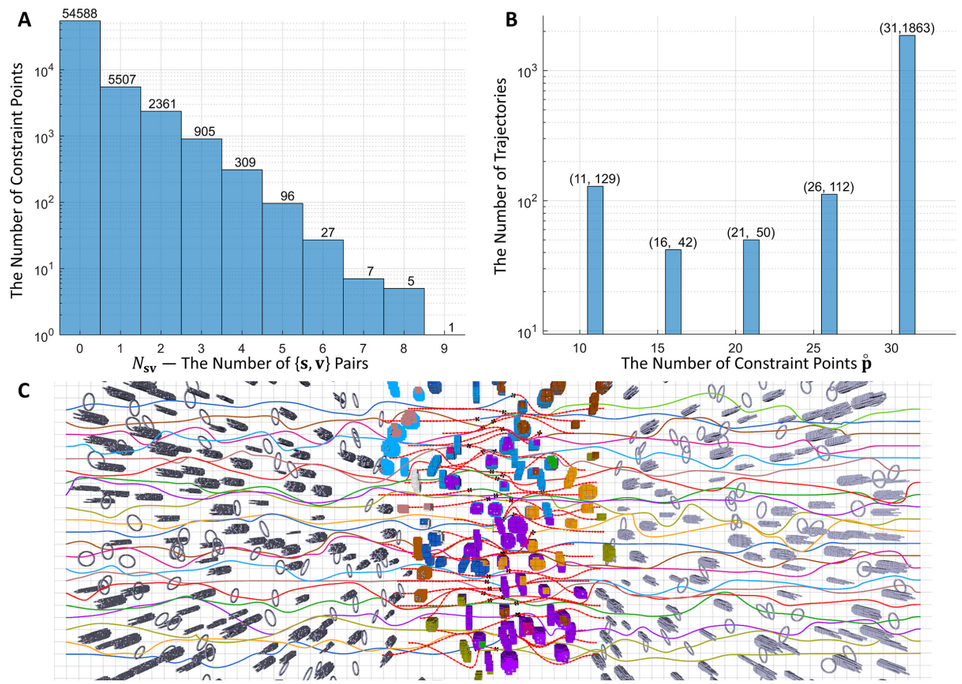
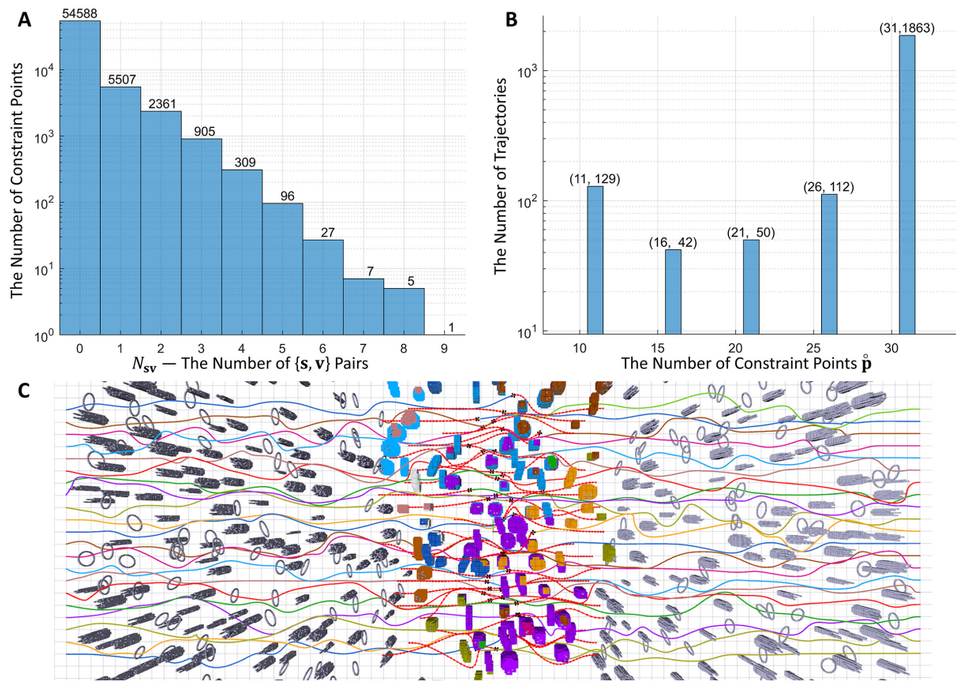


Fig. S16. Histograms of $N_{\mathbf{sv}}$ and the number of constraint points $\mathring{\mathbf{p}}$. (**A**) The statistic of $N_{\mathbf{sv}}$ on each constraint point. (**B**) The number of constraint points on each trajectory. (**C**) The test scenario.



图S16. 直方图 $N_{\mathbf{sv}}$ 以及约束点的数量 $\mathring{\mathbf{p}}$。 (**A**) $N_{\mathbf{sv}}$在每个约束点上的统计。 (**B**) 每条轨迹上的约束点数量。 (**C**) 测试场景。

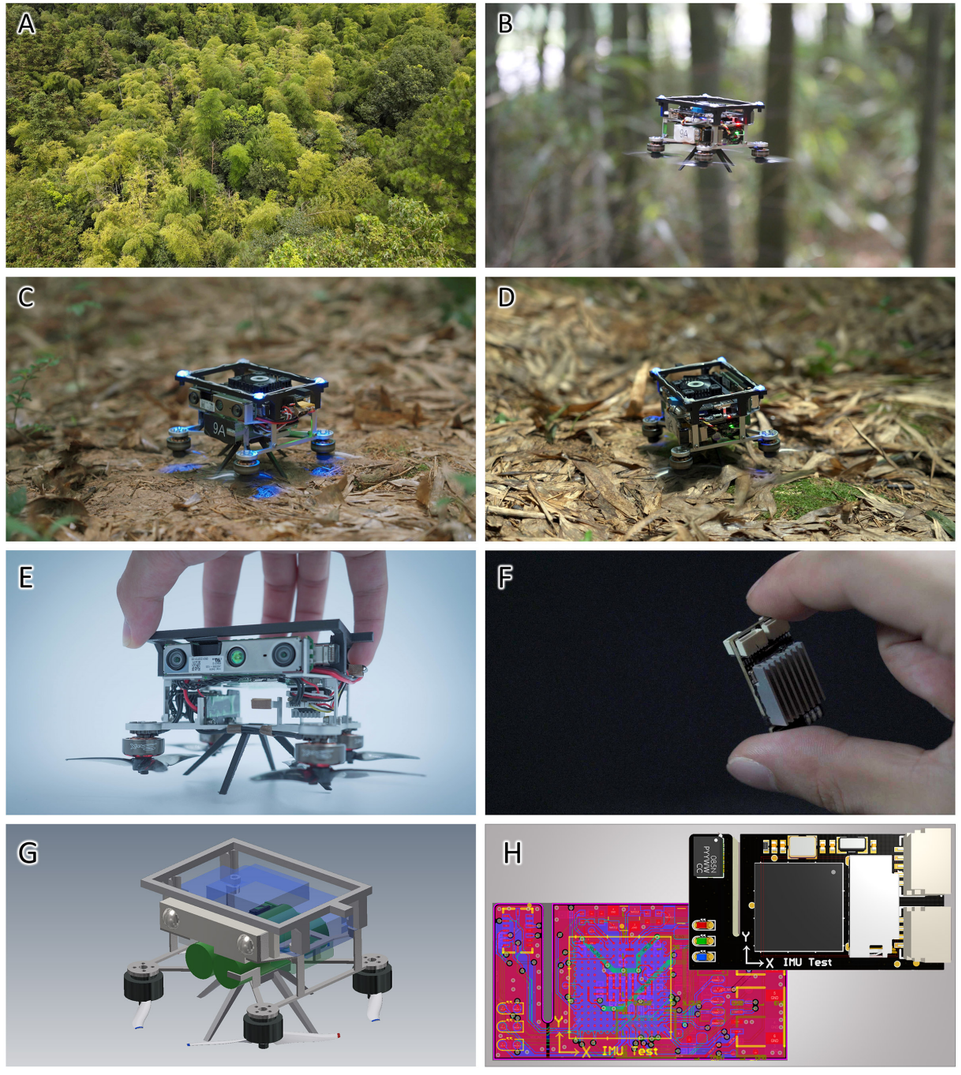
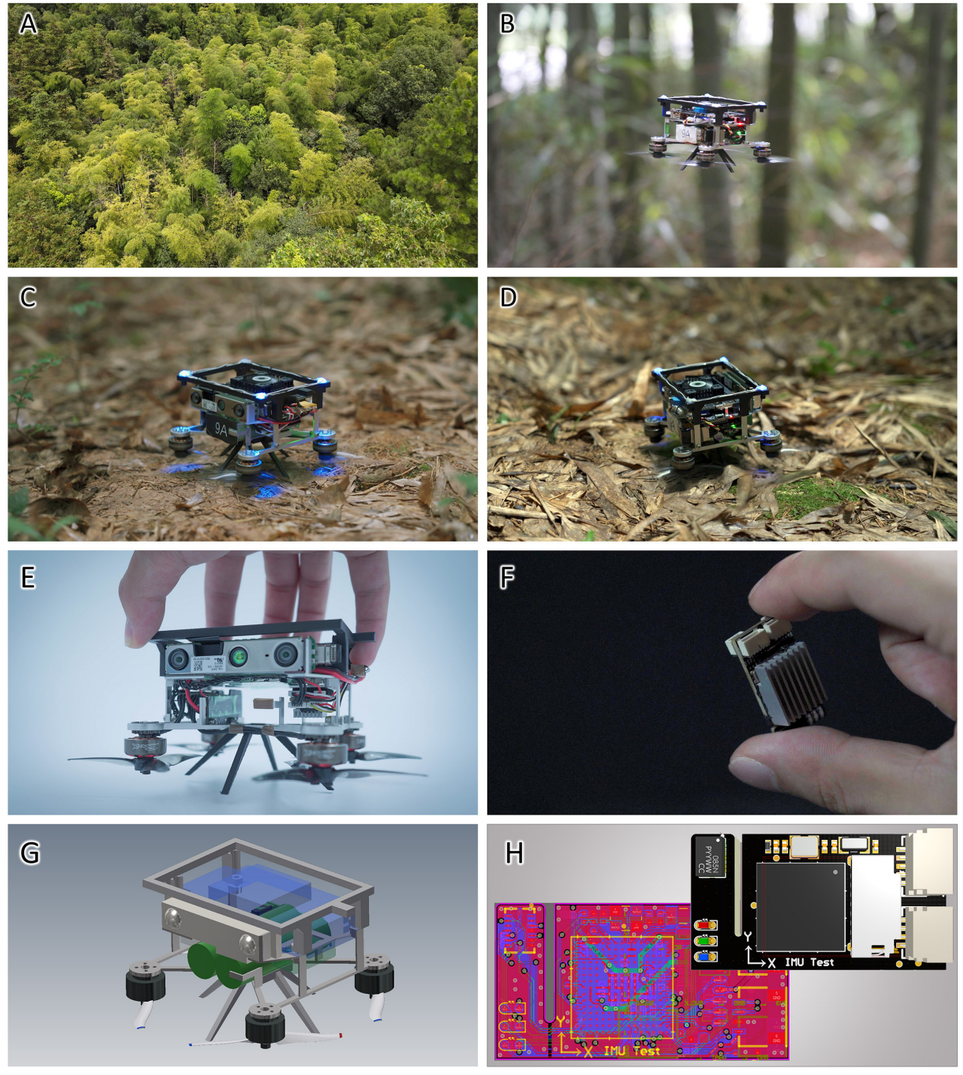


Fig. S17. More photographs of the environments and closeup views. (**A**) An aerial view of the bamboo forest where experiments are conducted. (**B-E**) Closeup views of the drone. (**F**) A miniature flight controller. (**G**) A mechanical model. (**H**) A PCB layout of the flight controller. Both (G) and (H) are released.



图S17. 更多环境照片和特写视图。 (**A**) 进行实验的竹林的鸟瞰图。 (**B-E**) 无人机的特写视图。 (**F**) 微型飞控器。 (**G**) 一个机械模型。 (**H**) 飞控器的PCB布局。G 和 H 均已发布。

Tables

Table S6. Parameter settings used throughout this work. Capital letters A–H indicate various scenarios: **A** includes the 10-/40-drone swarm flight, circle position swap, circle random goals, goal select and set, moving obstacle avoidance, scalability test on drone number, and benchmark with various obstacle densities: **B** is the formation flight simulation, **C** is the tracking simulation, **D** is the velocity scalability benchmark, **E** is the real flight in the bamboo forest, **F** is the real-world formation flight, **G** is the real-world intensive reciprocal avoidance evaluation, and **H** is the real-world swarm tracking. Parameter λ_u is defined in Section *Penalty Formulation*. *Formation weight* λ_{fo} and *Tracking weight* λ_{tr} are not explicitly defined elsewhere, but they follow the same naming conventions as other weights.

Parameters	**Scenarios**	**Values**
Trajectory piece number M	All	5
Number of constraint points per piece κ	All	5
Smoothness weight λ_s	All	1.0
Total time weight λ_t	All	10.0
Obstacle avoidance weight λ_o	All	10000.0
Swarm collision avoidance weight λ_w	All	10000.0
Dynamical feasibility weight λ_d	All	10000.0
Uniform distribution weight λ_u	All	10000.0
Formation weight λ_{fo}	B, F	100.0
Tracking weight λ_{tr}	C, H	100.0
Obstacle clearance $\mathcal{C}_o$	A-D	0.3m
	E-H	0.2m
Swarm clearance $\mathcal{C}_w$	A-D	0.5m
	E-H	0.4m
Maximum velocity v_m	A-C	1.5m/s
	E-H	1.0m/s
	D	1/2/3m/s
Maximum acceleration a_m	All	6m/s^2
Maximum jerk j_m	All	10m/s^2
Planning distance	All	7.5m
Emergency stop time threshold	All	1s

表格

表S6. 本研究中使用的参数设置。 大写字母A–H表示不同的场景：**A** 包括10/40无人机群飞行、圆形位置交换、圆形随机目标、目标选择与设定、避障移动、无人机数量的可扩展性测试，以及不同障碍密度的基准测试：**B** 是编队飞行仿真，**C** 是跟踪仿真，**D** 是速度可扩展性基准，**E** 是在竹林中的实飞，**F** 是实际环境中的编队飞行，**G** 是实际环境中的密集互避评估，**H** 是实际环境中的群体追踪。参数 λ_u 在章节惩罚函数中定义。编队权重 λ_{fo} 和 跟踪权重 λ_{tr} 在其他地方未明确定义，但它们遵循与其他权重相同的命名规则。

参数	**场景**	**数值**
轨迹片段数量 M	全部	5
每个片段的约束点数量 κ	全部	5
平滑度权重 λ_s	全部	1.0
总时间权重 λ_t	全部	10.0
避障权重 λ_o	全部	10000.0
群体避碰权重 λ_w	全部	10000.0
动力学可行性权重 λ_d	全部	10000.0
均匀分布权重 λ_u	全部	10000.0
形成权重 λ_{fo}	B, F	100.0
跟踪权重 λ_{tr}	C, H	100.0
障碍物清除 $\mathcal{C}_o$	A-D	0.3米
	E-H	0.2米
蜂群空域 $\mathcal{C}_w$	A-D	0.5米
	E-H	0.4米
最大速度 v_m	A-C	1.5米/秒
	E-H	1.0米/秒
	D	1/2/3米/秒
最大加速度 a_m	全部	6米每秒^2
最大冲击 jerk j_m	全部	10米每秒^2
规划距离	全部	7.5米
紧急停止时间阈值	全部	1秒

Table S7. Component weights.

Component	Weight (g)
Onboard computer	69.1
RealSense D430	19.0
FCU	5.3
UWB	4.7
Battery	101.4
ESC	4.5
Four motors & propellers	42.8
WiFi adapter	3.4
Drone frame	19.5
Power management unit	6.5
Screws, cables, etc.	17.3
Total	**293.5**

表S7. 组件重量。

组件	重量（g）
机载计算机	69.1
RealSense D430	19.0
FCU	5.3
超宽带 (UWB)	4.7
电池	101.4
ESC	4.5
四个电机与螺旋桨	42.8
WiFi适配器	3.4
无人机机架	19.5
电源管理单元	6.5
螺钉 电缆 等。	17.3
总计	**293.5**

38

Table S8. The values of sub-figure (1,1) in Fig. 7B.

Computing Time (MADER)			
Obstacle Spacing	**2.0**	**1.5**	**1.0**
Median	0.10650	0.10860	0.14150
Lower Quartile	0.06390	0.06190	0.07790
Upper Quartile	0.15550	0.16600	0.20580
Computing Time (EGO-Swarm)			
Obstacle Spacing	**2.0**	**1.5**	**1.0**
Median	0.00089	0.00120	0.00330
Lower Quartile	0.00050	0.00059	0.00079
Upper Quartile	0.00280	0.00370	0.01010
Computing Time (Proposed)			
Obstacle Spacing	**2.0**	**1.5**	**1.0**
Median	0.00035	0.00041	0.00063
Lower Quartile	0.00017	0.00020	0.00034
Upper Quartile	0.00063	0.00068	0.00097

表S8. 图7B中子图（1,1）的数值。

计算时间（MADER）			
障碍物间距	**2.0**	**1.5**	**1.0**
中位数	0.10650	0.10860	0.14150
下四分位数	0.06390	0.06190	0.07790
上四分位数	0.15550	0.16600	0.20580
计算时间（EGO-群体）			
障碍物间距	**2.0**	**1.5**	**1.0**
中位数	0.00089	0.00120	0.00330
下四分位数	0.00050	0.00059	0.00079
上四分位数	0.00280	0.00370	0.01010
计算时间（提出的方法）			
障碍物间距	**2.0**	**1.5**	**1.0**
中位数	0.00035	0.00041	0.00063
下四分位数	0.00017	0.00020	0.00034
上四分位数	0.00063	0.00068	0.00097

Table S9. The values of sub-figure (2,1) in Fig. 7B.

Computing Time (MADER)			
Velocity	**1.0**	**2.0**	**3.0**
Median	0.10650	0.10860	0.14150
Lower Quartile	0.06390	0.06190	0.07790
Upper Quartile	0.15550	0.16600	0.20580
Computing Time (EGO-Swarm)			
Velocity	**1.0**	**2.0**	**3.0**
Median	0.00081	0.00120	0.00210
Lower Quartile	0.00048	0.00059	0.00078
Upper Quartile	0.00240	0.00370	0.00620
Computing Time (Proposed)			
Velocity	**1.0**	**2.0**	**3.0**
Median	0.00043	0.00041	0.00029
Lower Quartile	0.00021	0.00020	0.00017
Upper Quartile	0.00078	0.00068	0.00065

表S9. 图7B中子图（2,1）的数值。

计算时间（MADER）			
速度	**1.0**	**2.0**	**3.0**
中位数	0.10650	0.10860	0.14150
下四分位数	0.06390	0.06190	0.07790
上四分位数	0.15550	0.16600	0.20580
计算时间（EGO-群体）			
速度	**1.0**	**2.0**	**3.0**
中位数	0.00081	0.00120	0.00210
下四分位数	0.00048	0.00059	0.00078
上四分位数	0.00240	0.00370	0.00620
计算时间（提出的方法）			
速度	**1.0**	**2.0**	**3.0**
中位数	0.00043	0.00041	0.00029
下四分位数	0.00021	0.00020	0.00017
上四分位数	0.00078	0.00068	0.00065

Table S10. The values of sub-figures (1,2) and (2,2) in Fig. 7B.

Minimum Clearance			
Obstacle Spacing	**2.0**	**1.5**	**1.0**
MADER	0.008	0.011	-0.184
EGO-Swarm	0.045	0.067	0.064
Proposed	0.086	0.065	0.058
Minimum Clearance			
Velocity	**1.0**	**2.0**	**3.0**
MADER	0.000	0.011	-0.086
EGO-Swarm	0.062	0.067	0.047
Proposed	0.068	0.065	0.064

表S10. 图7B中子图（1,2）和（2,2）的数值。

最小间隙			
障碍物间距	**2.0**	**1.5**	**1.0**
MADER	0.008	0.011	-0.184
EGO-群	0.045	0.067	0.064
建议	0.086	0.065	0.058
最小间隙			
速度	**1.0**	**2.0**	**3.0**
MADER	0.000	0.011	-0.086
EGO-群	0.062	0.067	0.047
提议	0.068	0.065	0.064

Table S11. The values of sub-figures (1,3) and (2,3) in Fig. 7B.

Control Effort				
Obstacle Spacing		**2.0**	**1.5**	**1.0**
MADER	**Average**	6608.10	7575.70	8922.20
	SD	4254.50	4707.70	4958.20
EGO-Swarm	**Average**	624.30	523.94	1133.30
	SD	346.29	362.08	811.70
Proposed	**Average**	189.03	246.42	465.18
	SD	98.87	129.27	183.72
Control Effort				
Velocity		**1.0**	**2.0**	**3.0**
MADER	**Average**	743.52	7575.70	19994.00
	SD	1271.70	4707.70	7195.70
EGO-Swarm	**Average**	13.26	523.94	4176.30
	SD	10.29	362.08	2945.20
Proposed	**Average**	70.64	246.42	456.18
	SD	36.14	129.27	207.19

表S11. 图7B中子图（1,3）和（2,3）的数值。

控制努力				
障碍物间距		**2.0**	**1.5**	**1.0**
MADER	**平均值**	6608.10	7575.70	8922.20
	标准差	4254.50	4707.70	4958.20
EGO-群体	**平均值**	624.30	523.94	1133.30
	标准差	346.29	362.08	811.70
提出的方法	**平均值**	189.03	246.42	465.18
	标准差	98.87	129.27	183.72
控制努力				
速度		**1.0**	**2.0**	**3.0**
MADER	**平均值**	743.52	7575.70	19994.00
	标准差	1271.70	4707.70	7195.70
EGO-群	**平均值**	13.26	523.94	4176.30
	标准差	10.29	362.08	2945.20
提出的	**平均值**	70.64	246.42	456.18
	标准差	36.14	129.27	207.19

42

Table S12. The values of sub-figure (1,4) in Fig. 7B.

Flight Time (MADER)			
Obstacle Spacing	**2.0**	**1.5**	**1.0**
Median	18.00	18.66	21.13
Lower Quartile	16.53	17.74	19.11
Upper Quartile	19.00	20.00	23.74
Flight Time (EGO-Swarm)			
Obstacle Spacing	**2.0**	**1.5**	**1.0**
Median	23.12	23.40	24.15
Lower Quartile	21.47	22.14	22.82
Upper Quartile	24.69	24.82	26.35
Flight Time (Proposed)			
Obstacle Spacing	**2.0**	**1.5**	**1.0**
Median	21.25	20.79	20.75
Lower Quartile	20.00	20.14	19.70
Upper Quartile	22.27	22.58	22.71

表S12. 图7B中子图（1,4）的数值。

飞行时间（MADER）			
障碍物间距	**2.0**	**1.5**	**1.0**
中位数	18.00	18.66	21.13
下四分位数	16.53	17.74	19.11
上四分位数	19.00	20.00	23.74
飞行时间（EGO-Swarm）			
障碍物间距	**2.0**	**1.5**	**1.0**
中位数	23.12	23.40	24.15
下四分位数	21.47	22.14	22.82
上四分位数	24.69	24.82	26.35
飞行时间（建议）			
障碍物间距	**2.0**	**1.5**	**1.0**
中位数	21.25	20.79	20.75
下四分位数	20.00	20.14	19.70
上四分位数	22.27	22.58	22.71

Table S13. The values of sub-figure (2,4) in Fig. 7B.

Flight Time (MADER)			
Velocity	**1.0**	**2.0**	**3.0**
Median	35.00	18.66	14.65
Lower Quartile	33.03	17.73	13.41
Upper Quartile	36.09	20.00	16.15
Flight Time (EGO-Swarm)			
Velocity	**1.0**	**2.0**	**3.0**
Median	47.96	23.40	15.58
Lower Quartile	45.51	22.14	14.76
Upper Quartile	51.48	24.82	16.60
Flight Time (Proposed)			
Velocity	**1.0**	**2.0**	**3.0**
Median	35.37	20.79	17.72
Lower Quartile	34.19	20.14	16.66
Upper Quartile	37.45	22.58	18.93

表S13. 图7B中子图（2,4）的数值。

飞行时间（MADER）			
速度	**1.0**	**2.0**	**3.0**
中位数	35.00	18.66	14.65
下四分位数	33.03	17.73	13.41
上四分位数	36.09	20.00	16.15
飞行时间（EGO-Swarm）			
速度	**1.0**	**2.0**	**3.0**
中位数	47.96	23.40	15.58
下四分位数	45.51	22.14	14.76
上四分位数	51.48	24.82	16.60
飞行时间（建议）			
速度	**1.0**	**2.0**	**3.0**
中位数	35.37	20.79	17.72
下四分位数	34.19	20.14	16.66
上四分位数	37.45	22.58	18.93